

Mental Models of Information: The Role of Attention and Memory*

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How do people form mental models, or *representations*, of decision environments? We theoretically and empirically show that attention and memory constraints generate predictable distortions depending on how the decision environment taxes cognitive resources. Suppose an agent faces a choice between assets. Constrained attention leads to favoring assets with larger payoff contrasts, while memory constraints lead to favoring assets with frequent payoff differences. Our framework has implications for preference identification and welfare: as we show, decision anomalies previously ascribed to preferences (probability weighting, experience effects) can arise from incorrect representations. It also provides a microfoundation for model misspecification.

Keywords: mental representation, behavioral economics, bounded rationality, model misspecification, beliefs, attention, memory, identification description-experience gap

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1 Introduction

How do people process information about a decision problem? In the standard framework people accurately internalize information about the choice environment and make decisions based on this correct mental representation and their preferences. We show that, when the choice environment is complex, limits on cognitive resources—specifically, memory and attention—lead to systematic distortions of the mental representation. Critically, these distortions vary in predictable ways based on how features of the decision environment tax cognitive resources.

Incorrect mental representations have important implications for the study of choice: they lead to an identification problem akin to the one highlighted by [Manski \(2004\)](#), who argues that in settings with partial information, many sets of preferences and subjective beliefs can be consistent with choice data.¹ Similarly, choice behavior may be consistent with many sets of preferences and mental representations—even when decision makers have full information about the objective distribution. Indeed, we show that representational distortions can generate decision anomalies that have previously been attributed to preferences, such as the four-fold pattern of risk. Thus, in environments where cognitive constraints are likely to bind, data on mental representations is crucial for the analysis and interpretation of choice data.

To see how cognitive constraints and the decision environment interact to distort the mental representation, consider a choice between two assets, Asset F and Asset U. Each asset has the *same* underlying distribution of returns (hereafter, outcomes), which depend on the state of the world, but differ in the outcome associated with each state. Specifically, Asset F outperforms Asset U in all but one state and Asset U outperforms Asset F by a lot in the remaining state (see [Fig. 1](#)). People observe information about the assets either simultaneously, where the outcome in each state is shown at the time of choice, or sequentially, where the outcomes are presented one state at a time.² In both cases they are exposed to the same information.³ They then make a decision between the two assets. When people have a correct mental representation, the way that they are exposed to information—simultaneously versus sequentially—should have no impact on choice.

How do limits on attention and memory distort people’s mental representations?

¹[Manski \(2004\)](#) argues that data on behavior can be rationalized by different sets of preferences and beliefs. This implies that the recovery of preferences from choice data requires either strong assumptions (e.g., rational expectations) or belief data.

²This distinction between simultaneous or sequential information reflects real-world differences in how people encounter information about their choice environment. In some cases, they have access to information about all the potential outcomes and their likelihoods simultaneously (e.g., price charts describing the historical returns of a stock, sports betting, casino gambling, public lotteries.) In other cases, people are exposed to a relevant distribution by sequentially sampling from it (e.g., the riskiness of a new business relationship or a stock is judged by observing and experiencing the relevant outcomes.)

³To keep information constant, the sequential case corresponds to sampling without replacement until all the outcomes associated with all states are observed.

Forming a mental representation requires encoding data on each asset’s outcomes in each state. When this dataset is sufficiently large—that is, there are many states (or assets)—it is representationally complex and processing it requires substantial attention and memory resources. The way that people are exposed to data will differentially tax these cognitive resources.⁴ When information is presented simultaneously—so that many objects are displayed at the same time—limited attention implies that not all objects in the data can be attended to equally. Models of salience predict that people will focus on states with bigger payoff contrasts, such as the state in which Asset U outperforms Asset F. Channeling attention to this salient state lead it to be overweighed in the mental representation—resulting in a mental representation that overestimates the expected value of Asset U relative to Asset F. Thus, salience-channeled attention to states with bigger payoff contrasts can cause people to favor prospects that outperform by a large margin, even if they most often underperform—choice behavior that looks like preferences that disproportionately weight outcomes of rare events.

When information is presented sequentially—so that a few objects are displayed at a time and then must be recalled at the time of choice—imperfect memory recall implies that not all objects can be correctly recalled. An infrequent event—such as the one state in which Asset U outperforms Asset F—is more likely to be misremembered.⁵ Forgetting the outcomes in this state means that it is no longer salient at the time of choice, which leads to a mental representation where Asset F dominates Asset U and its expected return is overestimated relative to Asset U. Imperfect recall can thus shift choices towards prospects that frequently outperform relative to prospects that infrequently outperform, even if the infrequent out-performance is by a large margin—choice behavior that looks like preferences that disproportionately discount outcomes of rare events.

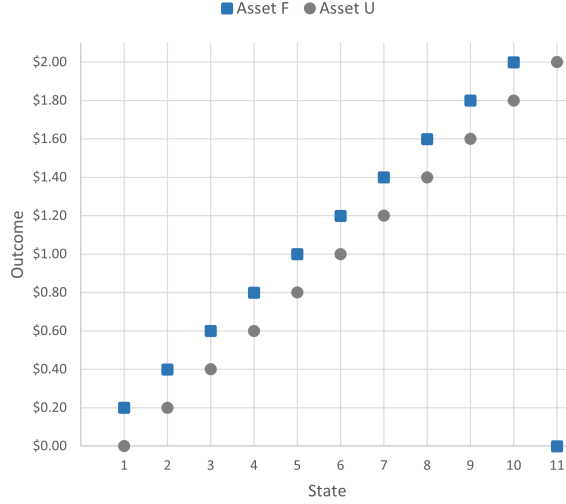
We tested these predictions on how constrained attention and imperfect memory distort people’s mental representations in an experiment that mirrors the setup described above. In our *Baseline* treatments, participants were exposed to information about Assets F and U either simultaneously or sequentially. We then elicited participants’ representations of the information environment—in terms of the perceived expected return and the subjective likelihoods of low versus high outcomes for each asset—and their choice between the assets.⁶ Since the information at the

⁴A recent literature in cognitive psychology and economics differentiates between representational and computational complexity as a function of what cognitive resources are taxed. The former—which is a function of how many objects one needs to consider to form an accurate mental representation—primarily taxes attention and memory resources while the latter taxes resources involved in controlled processing (Shenhav et al., 2017; Ba et al., 2025, 2022).

⁵As outlined formally in Section 2, imperfect recall implies that as the number of objects one needs to remember increases, the chances of remembering any one object decreases (Anderson et al., 1998).

⁶The order of elicitation was counterbalanced and one of the two was randomly selected to be incentivized for payment.

Figure 1: Distributions of Outcomes: Frequently-outperforming Asset F (blue) versus unlikely-outperforming Asset U (gray). All states are equally likely (1-in-11). The marginal outcome distributions of the two assets are identical.



time of elicitation was the same in both the simultaneous and sequential treatments, the standard model predicts no differences in mental representations or decisions. Additionally, because both assets have the same marginal distribution, correlation-insensitive models—where choice is independent of the correlation between the two assets (e.g., expected utility, cumulative prospect theory)—predict that participants will be indifferent between the assets in both treatments (Lanzani, 2022). Variation in the information exposure method thus serves as a conservative test for determining whether distorted mental representations impact choice behavior.

Participants had significantly biased mental representations that systematically differed with the information exposure method: when exposed to information simultaneously (sequentially), they overestimated the performance of Asset U (Asset F). Notably, the direction of the distortion *reversed* across exposure methods, suggesting that the way in which information was presented influenced how participants internalized it. Choices mirrored this difference in mental representations: participants selected Asset F 42% of the time when exposed to information simultaneously and 85% of the time when exposed to information sequentially. This represents a more than 40 percentage point gap in choices even though the distributions of the assets were identical and remained the same across exposure methods.

Imposing rational expectations to interpret this choice data would result in attributing the representational distortions to preferences—leading to an incorrect conclusion about the latter. In contrast, by eliciting the mental representation we are able to show that the observed choice behavior stems from distorted perceptions. Moreover, by comparing the elicited mental representation to the objective distributions, we can classify the mental representation as biased and therefore choices as mistakes. These conclusions have important implications for welfare.

Thus far, we have shown that distorted mental representations drive choice behavior and distortions are consistent with those predicted to arise from constrained attention and memory, but we have not provided a direct causal link between the latter. In order to establish this link, we ran additional treatments that measure attention and exogenously manipulate factors that influence attention and memory.

First, we measured attention in the simultaneous condition using standard tools from cognitive psychology (specifically, the Mouselab paradigm (Payne et al., 1988), eye-tracking (Yang and Krajbich, 2021), and a machine learning algorithm (Harel et al., 2006)). Indeed participants’ attention was drawn to the predicted most salient state (state 11 where Asset U outperforms Asset F by a large margin). Moreover, this channeled attention was predictive of choice: individuals whose attention was drawn to the most salient state were more likely to choose Asset U than individuals whose attention was drawn to other states.

Second, we manipulated factors that impact attention in the simultaneous exposure setting. Specifically, we shuffled the order of the states so that the outcomes in state 11 where Asset U outperformed Asset F were less distinct from nearby outcomes. If salience depends on departures from the local norm, as in our framework, then this reordering will reduce the salience of state 11.⁷ This change in salience completely eliminated the difference in participants’ perceived expected return of each asset. Choices followed suit: participants had a higher propensity to choose Asset F in this *No Salience Bias* treatment compared a higher propensity to choose Asset U in *Baseline Simultaneous*. The observed choice reversal provides evidence for salience-channeled attention as a driver of the overweighing of unlikely events, as has been observed in prior work (e.g., Kahneman and Tversky, 1979). It also rules out correlation-sensitive models based on regret (Loomes and Sugden, 1982) as an explanation and shows that payoff-based salience depends on payoffs in both the state itself and nearby states.⁸

Third, we manipulated factors that impact memory in the sequential exposure setting. In the *No Recall Bias* treatment, we decreased the scope for imperfect recall by asking participants to record the outcomes in each state while observing the sequential sample. Participants could consult this record at the time of choice.⁹ Reducing memory constraints eliminated the favoring of Asset F in participants’ mental representations: participants perceived both assets to have the same expected return, as in *No Salience Bias*. Choices followed suit: the propensity to choose Asset

⁷Indeed, measured attention verified that attention was no longer systematically drawn to state 11 over other states. The treatment kept the information exposure method and underlying properties of the assets (including the state-specific outcomes) constant compared with the *Baseline Simultaneous* treatment

⁸Prior theoretical work on payoff-based salience focused on contrasts within but not across states (Bordalo et al., 2012). Our results enrich this framework by taking the latter into account as well.

⁹The information exposure method and protocol were the same as in the *Baseline Sequential* treatment.

F decreased from the high level observed in *Baseline Sequential* to a level similar to that observed in *No Salience Bias* (roughly 60%)

By eliminating the bias in the mental representations, observing choices that favor Asset F in *No Recall Bias* and *No Salience Bias* isolate a preference for Asset F. Notably, the similarity in mental representations and choices between *No Recall Bias* and *No Salience Bias*—despite different exposure methods—suggests that the different representations and choices in the baseline treatments stemmed from the way that simultaneous versus sequential exposure differentially taxed attention and memory resources, as opposed to inherent differences between these exposure methods per se.

To further demonstrate how memory impacts the mental representation, we manipulated feature-based similarity—a critical factor governing memory recall (Kahana, 2012; Anderson et al., 1998). Memory recall functions through association: given a cue, an outcome’s probability of being correctly recalled increases with its similarity to the cue. When multiple outcomes share similar features, this creates the potential for *interference*: a person may incorrectly recall a different similar outcome. In a set of interference treatments, we added a third decoy asset (Asset D). Participants observed the outcome distribution for Asset D alongside Assets U and F but were told ahead of time that they could not choose Asset D. In one variation, Asset D had similar but payoff-irrelevant features to Asset U and in the other, it had similar features to Asset F. Asset D’s payoffs dominated both Assets U and F.

Interference predicts that the presence of the decoy Asset D will inflate the perceived expected return of the asset similar to it (U or F depending on treatment), since this similarity causes participants to sometimes incorrectly recall the higher outcomes of Asset D when forming beliefs about the similar asset. Our results match this prediction: in the *Interference Asset F* treatment, where the decoy was similar to Asset F, Asset F was perceived to have a higher expected value than Asset U; the reverse was true in the *Interference Asset U* treatment where the decoy was similar to Asset U. This difference in mental representations generated a shift in choices: Asset F was selected roughly 80% of the time in *Interference Asset F* and just over 30% of the time in *Interference Asset U*. This almost 50 percentage point shift in choices arose simply from varying the feature-based similarity of the decoy asset.¹⁰

The attention and memory manipulations outlined above varied factors associated with the respective cognitive constraints while holding the choice environment and sequential or simultaneous nature of information exposure constant. Our results thus suggest that neither a preference for low-probability events (e.g., probability weighting) nor some inherent aspect of being exposed to information simultaneously or sequentially (e.g., reinforcement learning) are responsible for the differences in mental representations and the choice gaps in the baseline treatments.

¹⁰The shift in choices resembles a “decoy effect” Tversky and Simonson (1993) but is conceptually different because the manipulated attributes were payoff-irrelevant.

To show how our framework can rationalize well-documented choice anomalies, we apply it to a setting used to demonstrate the four-fold pattern of risk. This pattern corresponds to risk aversion over high probability gains and low probability losses, but risk seeking over low probability gains and high probability losses. It is typically attributed to preference primitives that affect how people weight probabilities (Tversky and Kahneman, 1992). In our framework, salience-channeled attention can lead to a mental representation that generates similar choice behavior. To isolate the role of constrained attention in driving the four-fold pattern of risk, we exogenously manipulate salience in settings with either a low or high probability of a gain. We first replicate risk seeking and risk aversion in a baseline context. We then introduce a decoy asset whose payoffs increase the salience of the state that has either a low-probability gain or a low-probability no-gain (i.e., the asset has a high probability of a gain). Consistent with the predicted impact of constrained attention, the introduction of the decoy asset exacerbates risk seeking (risk aversion) in the treatment with a low probability of a gain (no gain).

Finally, we show that our framework can also be applied to settings with non-risky choice. When the choice environment requires a person to consider a large set of objects to evaluate a decision, memory and attentional distortions can similarly impact the mental representation and hence, choice. To demonstrate this application, we run versions of our baseline treatments where participants are paid according to the average outcome of their chosen asset, rather than the outcome in a randomly drawn state. Our results replicate both the gap in mental representations and decisions across both exposure treatments. These results contextualize prior work on complexity in risky choice (Oprea, 2024) as being driven at least in part by the interaction between representational complexity and constrained attention.

Our framework builds on the economics and psychology literatures on limited attention, salience, and bounded memory (Bordalo et al., 2012; Li and Camerer, 2022; Bruce and Tsotsos, 2009; Anderson et al., 1998; Bordalo et al., 2022a). The distorted mental representation can be thought of as a *misspecified model* (Bohren and Hauser, 2024). Our model of how attention and memory distort the mental representation thus provides a foundation for model misspecification.¹¹ Our framework also provides a unified explanation for several well-documented choice anomalies, including the four-fold pattern of risk discussed above and the “description-experience” gap in risky choice, where people undervalue low-likelihood large-payoff prospects when learning about the distribution by experiencing returns and overvalue them when directly observing the distribution (Hertwig et al., 2004). By deriving these phenomena from first principles of bounded cognition, we can trace seemingly distinct behaviors back to the same underlying cognitive mechanisms. See Section 7 for further discusses this

¹¹In a similar vein, Ba et al. (2022) show how attention and processing constraints lead to model misspecification in an inference setting.

connection and other related work.

The rest of the paper proceeds as follows. [Section 2](#) outlines a model of constrained attention and memory and illustrates how these constraints interact with the information exposure method. [Section 3](#) introduces our experimental design. [Section 4](#) presents the results on mental representations and decisions. [Section 5](#) measures and manipulates attention and recall to identify these cognitive constraints as the mechanisms driving our results. [Section 6](#) shows how our framework can be used to rationalize the four-fold pattern of risk and presents results on non-risky choice. [Section 7](#) reviews the related literature and [Section 8](#) concludes with a discussion of the implications of our results.

2 Theoretical Framework

We present a framework for how an agent with bounded memory and limited attention mentally represents information about a risky choice. We derive predictions on how the interaction between these cognitive constraints and the way in which agents are presented with information about the decision environment impact this mental representation and choice. Further details and all proofs are in [Appendix A](#).

2.1 A Cognitive Model of Mental Representation and Choice

Decision Problem. An agent faces a choice between $N \geq 2$ assets. Each asset A^i specifies an outcome $x \in \mathbb{R}$ in each of $K > 1$ states $\omega \in \Omega$. Let $x^i : \Omega \rightarrow \mathbb{R}$ denote the mapping from states to outcomes, such that $x^i(\omega)$ is the outcome in state ω . The state is drawn according to $\pi \in \Delta(\Omega)$, where $\pi(\omega)$ denotes the probability of state ω .¹²

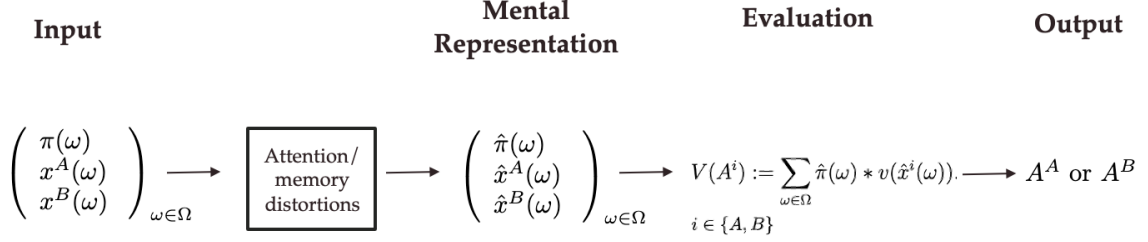
The agent uses a two-step procedure to make her choice. First, she forms a *mental representation*—her mental model—of the decision environment. Here, a mental representation consists of a perceived outcome mapping for each asset and a perceived distribution over states. Cognitive constraints may lead her mental representation to differ from the objective environment, as we model below. Next, she uses her mental representation to compute the expected utility of each asset given strictly increasing utility function $v : \mathbb{R} \rightarrow \mathbb{R}$ over outcomes. She chooses the asset with the highest expected utility. An illustration of this process is presented in [Fig. 2](#).

Forming a Mental Representation. Forming a mental representation of the decision environment requires internalizing $KN + K - 1$ pieces of data: the K different outcomes for each of the N assets and the $K - 1$ elements of the probability distribution over states. The size of the dataset captures the *representational* complexity of the decision environment—that is, the number of objects that need to be considered in order to form an accurate mental representation.¹³

¹² $\Delta(Z)$ denotes the set of all probability distributions on set Z .

¹³Representational complexity is distinct from other forms of complexity because it primarily taxes attentional and memory resources ([Shenhav et al., 2017](#); [Ba et al., 2022](#)). [Panichello and](#)

Figure 2: Decision Problem: Choose between Assets A and B.



When the decision environment is representationally complex—that is, there are many states or assets—the timing and order in which agents are exposed to the data differentially taxes memory and attentional resources, and therefore, how constraints on these resources distort the mental representation. For example, the timing of exposure—e.g., presenting the data beforehand or at the time of choice—will impact recall, while order of data exposure—e.g., presenting the assets side-by-side versus independently—will impact the salience of payoff contrasts between two assets and therefore how attention is directed.

We consider two variations on the timing of data exposure. In the *simultaneous information* setting, the agent observes a figure depicting the outcome distribution for each asset, mimicking settings where an agent is shown aggregate statistics about an asset before making a decision. This setting taxes attentional resources, as the agent is exposed to all data simultaneously and needs to determine how to direct attention to each piece of data in order to process it. It does not tax memory resources, as recall is not required; all data is available at the time of choice. In the *sequential information* setting, the agent observes the outcomes for all assets one state at a time, mimicking settings where an agent learns about an asset through sequential sampling. Importantly, the agent samples without replacement until all states are displayed, so that at the time of choice the agent has observed the same data in both variations. The sequential setting taxes memory resources, as it requires recalling the outcome mapping for each asset at the time of choice. We also consider several variations on the order of data exposure, as discussed in [Prediction 1](#) and [Section 5.2](#).

We next describe how the agent recalls and attends to data about the decision environment, and how this process interacts with the timing and ordering of data exposure.

Perceived Outcomes. We first consider what outcome mapping an agent has available at the time of choice. We refer to this mapping as her perceived outcome mapping, denoted by $\hat{x}^i : \Omega \rightarrow \mathbb{R}$ for asset A^i and $\hat{X} := (\hat{x}^1, \dots, \hat{x}^N)^T$ for all assets.

In the sequential information setting, the agent must recall the outcome each asset

Buschman (2021); Oberauer (2019) show how attention and memory play a critical role in forming mental representations.

assigns to each state. Memory recall depends on *memory load*—which is increasing in the number of outcomes necessary to recall—and *interference*—the misremembering of outcomes due to similarity of other assets (Bordalo et al., 2020; Anderson et al., 1998). Given a set of assets $\mathcal{A}$, define similarity function $S : \mathcal{A}^2 \rightarrow [0, 1]$, where $S(A, A')$ measures the similarity of A to A' for any assets $A, A' \in \mathcal{A}$ and values closer to one denote higher similarity. Assume similarity is symmetric, $S(A, A') = S(A', A)$.¹⁴ Let $s^{ij} := S(A^i, A^j)$ denote asset A^i 's similarity to asset A^j and $s^i := \{s^{ij}\}_{j \neq i}$ denote asset A^i 's corresponding vector of similarities to assets $j \neq i$. The recalled outcome $\hat{x}^i(\omega)$ for asset A^i in state ω depends on the number of outcomes necessary to recall, NK , and the similarity with other assets, s^i . In particular, the agent correctly recalls an outcome, $\hat{x}^i(\omega) = x^i(\omega)$, with probability

$$r(NK, s^i) := \min \left\{ 1, \frac{\alpha}{NK(1 + \sum_{j \neq i} s^{ij})} \right\},$$

where $\alpha > 0$ parameterizes the threshold number of outcomes and level of similarity at which the agent begins to misremember outcomes with positive probability (higher α means the threshold is higher). With probability $1 - r(NK, s^i)$, the agent incorrectly recalls an outcome; instead, she recalls the average between the correct outcome and the outcome of the most similar asset A^k : $\hat{x}^i(\omega) = (x^i(\omega) + x^k(\omega))/2$ where A^k is the asset that maximizes $\max_{j \neq i} S(A^i, A^j)$.¹⁵ Thus, the probability of correct recall decreases as representational complexity increases—which increases the memory load—or an asset grows more similar to other assets—which strengthens interference. We maintain the assumption that the recall process is independent across states.

In the simultaneous information setting, the outcome mappings are displayed at the time of choice. Therefore, there is no need for recall: $\hat{x}^i(\omega) = x^i(\omega)$ for all $\omega \in \Omega$ and all i .

Subjective State Distribution. We next consider how the perceived outcome mapping interacts with salience-channeled attention to determine how the agent mentally represents the probability of each state, which we refer to as her subjective state distribution and denote by $\hat{\pi} \in \Delta(\Omega)$.

When faced with a sufficiently large number of states, limited attention is channeled towards some states over others as a function of their salience.¹⁶ In our setting,

¹⁴Modern models of similarity (e.g., the Generalized Context Model) specify similarity as being symmetric (Nosofsky, 1986). Earlier approaches considered asymmetric similarity functions (Tversky, 1977).

¹⁵We assume there is a unique maximizer; this is satisfied in the empirical tests of the model.

¹⁶Research has shown that an agent can attend to a limited number of objects at a time—typically 3 or 4 (Oberauer et al., 2016; Luck and Vogel, 1997; Loewenstein and Wojtowicz, 2023; Bays et al., 2011). See Brandstätter and Körner (2014); Spitmaan et al. (2019); Bordalo et al. (2022b) for research on how salience channels attention.

key features that determine a state’s salience include payoffs in that state (Bordalo et al., 2013) and visual stimuli (Frydman and Mormann, 2016; Bose et al., 2022)—particularly, payoff contrast with nearby states.¹⁷ For brevity, in this section we take the salience ranking of states as a primitive of a given data exposure setting. Namely, let $k = (k(\omega_1), k(\omega_2), \dots, k(\omega_K))$ denote the ranking of each state by its relative salience.¹⁸ In Appendix A we formalize how salience depends on the payoff distribution and ordering in which these payoffs are displayed to provide a foundation for this salience ranking.¹⁹

The probability assigned to state ω is distorted proportional to its salience:

$$\hat{\pi}(\omega) := \frac{\pi(\omega)\delta^{k(\omega)}}{\sum_{\omega' \in \Omega} \pi(\omega')\delta^{k(\omega')}}, \quad (1)$$

where $\delta \in (0, 1]$ parameterizes the extent of the attentional constraint, with $\delta = 1$ corresponding to no attentional constraint and lower δ corresponding to more constrained attention. This salience distortion generates a mental representation that overweighs more salient states and underweighs less salient states relative to the objective distribution.²⁰ When $\delta = 1$, the subjective and objective distributions coincide, and when $\delta \approx 0$ the subjective distribution assigns all probability to the most salient state.

Evaluating Assets. Given a mental representation with perceived outcome mapping $\hat{X}$ and subjective state distribution $\hat{\pi}$, the perceived expected utility of asset A^i is its expected utility with respect to this mental representation,

$$V(A^i) := \sum_{\omega \in \Omega} \hat{\pi}(\omega) * v(\hat{x}^i(\omega)). \quad (2)$$

The agent chooses asset A^i over asset A^j if it has a higher perceived expected utility, $V(A^i) > V(A^j)$.²¹ Note that when there is imperfect recall, the recalled outcome mapping for an asset, and hence, its perceived expected utility, will be random. Thus in our predictions about choice, we focus on the probability that the agent

¹⁷For ‘nearby states’ we focus on states that are visually displayed adjacent to the given state.

¹⁸The salience of an object—hence, the salience ranking—can be predicted using algorithms (Bose et al., 2022; Li and Camerer, 2022; Bruce and Tsotsos, 2009) such as the Graph-Based Visual Saliency model (Harel et al., 2006). As such, salience can be viewed as an empirically measurable primitive—akin to risk aversion. We use this technique to validate our assumptions on the salience ranking of states in our experimental studies.

¹⁹Specifically, we combine payoff-based and visual salience by generalizing the payoff-based salience model used in Bordalo et al. (2013) to incorporate payoff contrast with nearby states.

²⁰Frydman and Mormann (2016) finds that states that are more salient are overweighed in the decision-process compared to states that are less salient. Ba et al. (2022) also provides empirical evidence on the importance of state-based salience in the domain of belief-updating.

²¹While we adopt expected utility in the evaluation stage, our framework can be adapted to alternative evaluation procedures such as non-expected utility (Machina, 1989) and cognitive noise (Woodford, 2020). See Section 5.4 for a discussion of how identifying the mental representation aids in the recovery of such preferences.

chooses each asset, i.e., $Pr(V(A^i) > V(A^j))$, where the probability is with respect to the recall process. As a benchmark, let $V^R(A^i) := \sum_{\omega \in \Omega} \pi(\omega) * v(x^i(\omega))$ denote the expected utility of asset A^i under the correct mental representation.

We also derive and test predictions on the perceived expected value of an asset. Analogous to perceived expected utility, the perceived expected value of asset A^i given perceived outcome mapping $\hat{X}$ and subjective state distribution $\hat{\pi}$ is

$$\hat{E}(A^i) := \sum_{\omega \in \Omega} \hat{\pi}(\omega) * \hat{x}^i(\omega). \quad (3)$$

Similarly, let $E(A^i) := \sum_{\omega \in \Omega} \pi(\omega) * x^i(\omega)$ denote the expected value of A^i under the correct mental representation.

Discussion. Our framework builds on several well-established aspects of limited attention and imperfect memory, such as the impact of salience on attribute weighting in judgment and choice as well as the impact of similarity and interference on memory recall. The model aims to formalize these aspects within a unified framework in order to explore the impact of the data exposure method on decision-making.

2.2 Predictions about Asset Valuations and Choice

We consider a choice between two assets, denoted A^U and A^F , that have the same marginal distributions over outcomes. Let $\mathcal{X} = (x_1, \dots, x_K)$ denote a set of outcomes ordered from lowest to highest. The outcome mapping is as follows. For $k = 1, \dots, K-1$, Asset A^U returns outcome x_k in state ω_k and Asset A^F returns outcome x_{k+1} . For state ω_K , Asset A^U returns outcome x_K and Asset A^F returns outcome x_1 . Thus A^F frequently outperforms A^U by a relatively small margin: it has a higher outcome in all but one state ω_K . Asset A^U infrequently outperforms A^F by a large margin: it has a higher outcome in state ω_K and the difference between outcomes is at a maximum across states. Each state is equally likely, $\pi(\omega) = 1/K$ for all $\omega \in \Omega$. Hence, the marginal distribution over outcomes and the objective expected utility are the same for both assets.

Our first prediction is on the evaluation of A^U and A^F in the simultaneous information setting, which we refer to as *Baseline Simultaneous*. Suppose states are displayed in order $(\omega_1, \dots, \omega_K)$ from the lowest payoff for asset A^U to the highest. In this setting, the agent has a correct perceived outcome mapping but overweighs more salient states. State ω_K is the most salient since it has the largest payoff contrast between assets and this payoff contrast is large compared to payoffs in the adjacent state ω_{K-1} . Note that since the perceived outcome mapping is deterministic (there is no probabilistic misremembering), so are perceived expected utility and choice.

Prediction 1 (Baseline Simultaneous). *Consider the simultaneous information setting with states ordered from the lowest to highest outcome for A^U and suppose the agent faces attentional constraints, $\delta < 1$. Then asset A^U has a higher perceived*

expected value and perceived expected utility, $\hat{E}(A^U) > \hat{E}(A^F)$ and $V(A^U) > V(A^F)$. Therefore, an expected utility maximizer chooses A^U over A^F .

This result follows from the observation that overweighing the likelihood of state ω_K inflates the perceived expected value—and therefore, expected utility—of A^U relative to A^F , since A^U has a higher payoff than A^F in state ω_K .

Our second prediction is on the evaluation of A^U and A^F in the sequential information setting, which we refer to as *Baseline Sequential*. In this setting, the agent may have an incorrect perceived outcome distribution due to misremembering outcomes. Moreover, recalled outcomes, and hence, perceived expected utility and choice, are stochastic.

Prediction 2 (Baseline Sequential). *Consider the sequential information setting with any ordering over states and suppose $\alpha/(1+s^{UF}) > \ln 2$. Then there exists a number of states $\bar{K}$ such that for $K > \bar{K}$, the agent is more likely to perceive that A^F has a higher expected return and expected utility, $\Pr(\hat{E}(A^F) > \hat{E}(A^U)) > 1/2$ and $\Pr(V(A^F) > V(A^U)) > 1/2$. Therefore, an expected utility maximizer chooses A^F with strictly higher probability than A^U .*

Asset A^U outperforms A^F in one state while A^F outperforms A^U in the remaining states. Thus, if the agent incorrectly recalls the outcomes in state ω_K where A^U outperforms A^F and correctly recalls at least one other outcome, then she will believe A^F has a higher expected utility than A^U —regardless of how salience impacts $\hat{\pi}$, and hence, regardless of the order in which states are displayed. The chance that the agent incorrectly recalls the outcomes in state ω_K and remembers at least one other outcome increases with the number of states, provided that the agent does not become too likely to forget everything (i.e., provided $\alpha/(1+s^{UF})$ is not very low).²²

Taken together, the first two predictions generate a choice reversal: A^U is chosen with higher probability than A^F in *Baseline Sequential* and A^F is chosen with higher probability than A^U in *Baseline Simultaneous*.

Our third prediction is on how interference impacts the evaluation of A^U and A^F in the sequential information setting. Suppose there is a third ‘decoy’ asset A^D that dominates A^U and A^F , in that it yields a higher payoff than either asset in all states, $x^D(\omega) > \max\{x^U(\omega), x^F(\omega)\}$ for all $\omega \in \Omega$. In the *Interference Asset U* environment, A^U is most similar to A^D and A^F is most similar to A^U : $S(A^U, A^D) > S(A^U, A^F) > S(A^F, A^D)$. In the *Interference Asset F* environment, A^F is most similar to A^D and A^U is most similar to A^F : $S(A^F, A^D) > S(A^U, A^F) > S(A^U, A^D)$. In both environments, the agent is presented with data about A^D alongside A^U and A^F in each state but still faces a choice between A^U and A^F —she does not have the

²²Specifically, the parameter constraint $\alpha/(1+s^{UF}) > \ln 2 \approx 0.69$ implies that in the case of two assets and one state ($N = 2, K = 1$), the agent correctly recalls each outcome with probability at least $\ln 2/2 \approx 0.35$. Thus, the constraint is mild.

option to choose A^D . We compare mental representations and choice in these two interference environments to the *Baseline Sequential* environment with no A^D .

Prediction 3 (Interference Sequential). *Consider the sequential information setting with any ordering over states and decoy Asset D. There exists a number of states $\bar{K}$ such that for $K > \bar{K}$:*

1. *The probabilities that the perceived expected value and perceived expected utility of A^U exceed that of A^F are higher in Interference Asset U than in Baseline Sequential, $Pr_{IU}(\hat{E}(A^U) > \hat{E}(A^F)) > Pr_S(\hat{E}(A^U) > \hat{E}(A^F))$ and $Pr_{IU}(V(A^U) > V(A^F)) > Pr_S(V(A^U) > V(A^F))$. Thus, an expected utility maximizer chooses A^U with strictly higher probability in Interference Asset U than in Baseline Sequential.*
2. *The probabilities that the perceived expected value and perceived expected utility of A^U exceed that of A^F are lower in Interference Asset F than in Baseline Sequential, $Pr_{IF}(\hat{E}(A^U) > \hat{E}(A^F)) < Pr_S(\hat{E}(A^U) > \hat{E}(A^F))$ and $Pr_{IF}(V(A^U) > V(A^F)) < Pr_S(V(A^U) > V(A^F))$. Thus, provided she does not choose A^U with probability zero in Baseline Sequential, an expected utility maximizer chooses A^U with lower probability in Interference Asset F than in Baseline Sequential.*

This prediction stems from the interference property that underlies models of memory and impacts the recall process. When an outcome is misremembered, it is replaced by the outcome associated with the most similar asset. Thus, when A^U is more similar to A^D than to A^F ($S(A^U, A^D) > S(A^U, A^F)$), its recalled outcomes are influenced by A^D instead of A^F . Since the outcomes for A^D are higher than either asset, this inflates the perceived outcomes of A^U . Thus, misremembering an outcome from A^U increases its perceived expected value. Since A^D is more similar to A^U than to A^F ($S(A^U, A^D) > S(A^F, A^D)$), its presence also increases the probability that an outcome from A^U is misremembered relative to the change in the probability that an outcome from A^F is misremembered. Together these forces lead to a higher probability that the perceived expected value for A^U exceeds that for A^F when A^D is present, and hence, a higher probability that A^U is chosen.

3 Experiment

Next we empirically test how the information exposure method impacts agent's mental representations and choices. Here we describe our *Baseline* experimental paradigm.²³ We recruited participants ($N = 785$) using the Prolific crowdsourcing platform. Participants were restricted to locations from the US and UK and balanced on gender. Summary statistics for participants are reported in Table D.1 in the Internet Appendix.

²³The experiment was pre-registered on AsPredicted.org, see [this link](#). Complete instructions are reported in a supplementary document provided under [this link](#).

3.1 Design

Participants made two rounds of decisions. In each round, they were first presented with information about the joint distribution of two risky assets. Each asset specified an outcome ranging from \$0 to \$2 in either 11 states, as shown in Fig. 1, or 51 states. Both assets had the same marginal distribution over outcomes, which was discrete uniform, and an expected return of \$1.²⁴ The structure of dependence was asymmetric: Asset F slightly outperformed Asset U in all but one state and underperformed by a large margin in the final state.²⁵

Mirroring the theoretical framework, participants were exposed to this information either simultaneously or sequentially, as described in more detail below. Participants then reported their mental representation of the information environment—specifically, the expected return of each asset and the probabilities of receiving an outcome less than \$1 (the mean/median), equal to \$1, or greater than \$1—and made a decision between the two assets. Whether the 11-state or 51-state round was shown first and whether the choice or mental representation was elicited first was randomized across participants. Whether Asset U or F was displayed first in the choice elicitation was also randomized across participants and rounds. We also elicited participants’ strength of preference, which included the option to indicate indifference in the choice between assets.

Prior to making these decisions, participants first received introductory instructions. These included a description of the upcoming tasks, the type of information they would have access to, as well as an explanation of how they would be paid. Specifically, they were told that one of the two rounds would be randomly selected for the bonus payment at the end of the experiment. Within that round, it was then randomly selected whether the participant would be paid according to their choice of asset—based on a draw from the selected asset’s distribution—or according to their elicited representation—a bonus of \$2 if a randomly selected belief (e.g., the average of Asset F’s outcomes) was within ± 0.03 of the statistically correct value.²⁶ Including a base fee, the average hourly salary for *Baseline* was more than \$13. Participants needed to pass a comprehension test to continue to the first round. The experiment concluded after the second round with some final questions (e.g., self-assessed risk aversion). The experiment’s overall structure is illustrated in Fig. C.1.

3.2 Varying the Information Exposure Method

In the *Baseline Simultaneous* condition, participants were exposed to data on the outcomes of both assets in all states and the probability of each state (uniform)

²⁴This design was inspired by the paradigm used in Paterson and Diekmann (1988) to study violations of Expected Utility axioms.

²⁵The participants’ instructions referred to Asset U and F as Asset A and B, respectively.

²⁶Danz et al. (2022) show that the binarized scoring rule results in bias in elicited representations and higher error rates. They argue that incentivizing truthful reporting using belief quantiles, which we use here, results in more truthful reporting.

through a figure akin to Fig. 1. The initial instructions outlined how the data would be presented without revealing the actual distributions. When being exposed to the information, participants received an explanation of the data in a written statement as well. The statement mentions all joint outcomes and their probabilities, thus fully describing the joint outcome distribution.²⁷ Notably, participants could continue to view the figure when reporting their representations and choices, removing the need to recall the distribution.

In the *Baseline Sequential* condition, participants sequentially sampled each potential state without replacement until each state was drawn once. When a state was drawn, participants were shown a bar chart and written numbers describing the outcome for each asset in this state, as in Fig. 3. In the initial instructions, participants were told that they would see the outcomes of each potential state and that each state was equally likely to be realized; they drew 11 (51) outcome pairs in the 11-state (51-state) round. Including every state in *Baseline Sequential* ensured that at the time of choice, participants had been exposed to the same information about the relevant distributions as those in *Baseline Simultaneous*.²⁸ Each sample was randomly ordered. Participants chose how long to view the outcomes of each state and when to move on by clicking “Next.” There was no option to go back. We collected viewing times for each state, which we use to analyze attention effects in Section 4.3. Participants did not have access to any information about the asset distributions when reporting their representations and choices.

We varied the method of exposure between subjects; participants were evenly divided between methods. *Baseline Simultaneous* was designed to test the role of salience-channeled attention in distorting mental representations, as outlined in Prediction 1. *Baseline Sequential* was designed to test the impact of imperfect recall in distorting mental representations, as outlined in Prediction 2.

3.3 Discussion of Design

There are several noteworthy aspects of this design.

First, eliciting the mental representation—the expected value and belief distribution over objective outcomes—allows us to identify and classify perceptions as *mistakes*. Interpreting seemingly-anomalous choice behavior typically requires one to contend with the possibility that decisions reflect underlying preferences rather than systematic mistakes. In contrast, comparing elicited mental representations to the objective information environment allows not only for the identification of the

²⁷See Fig. C.2 for the *Baseline Simultaneous* instructions including the verbal statement. We used two variations of the figure to display the asset distributions: one that was identical to Fig. 1 and the other had an additional visual cue around the final state (a box) as a way to further enhance the salience of the state. There was no significant difference between conditions in either the elicited representations or the choice data, suggesting that the departure from local norm was sufficient to channel attention to that state; we pool both versions when presenting the results.

²⁸This sampling design was inspired by the experiment in Ungeheuer and Weber (2023) who study the impact of outperformance on investment choice.

Figure 3: Information display in *Baseline Sequential*.



source bias in choice behavior (distorted perceptions not preferences), but also for classification of this behavior as a mistake. This has important consequences for interpreting welfare.

Second, exposing participants to the full sample in *Baseline Sequential* ensures that participants have been exposed to the same information at the time of choice in both conditions. This feature eliminates sampling error as a potential explanation for the description-experience choice gap (Fox and Hadar, 2006; Cubitt et al., 2022).

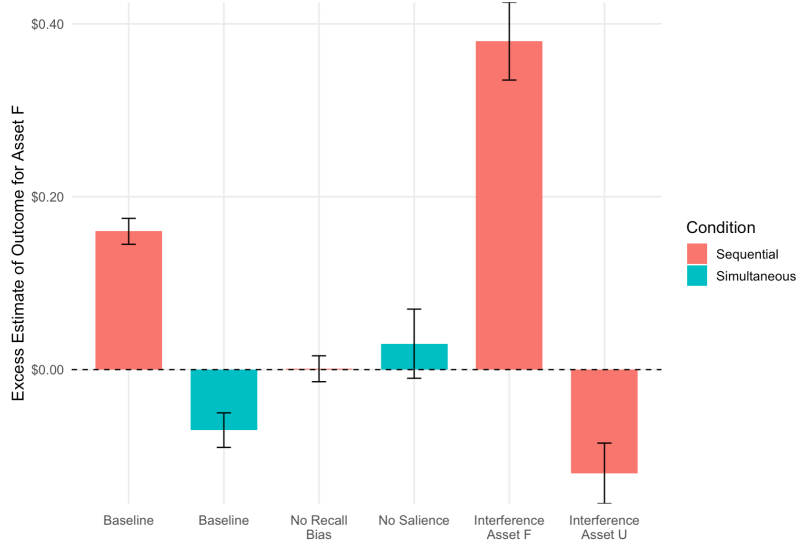
Third, displaying the outcomes of each state graphically in both treatments and selecting a uniform distribution over states rules out the “mere-presentation effect” as a potential explanation for our results (Erev et al., 2008).²⁹ The same design choice also rules out the “unpacking-and-repacking” explanation for our results (de Palma et al., 2014), which relies on a difference in emphasis on unlikely outcomes between simultaneous and sequential displays of information.

Fourth, both assets have the same marginal distribution by design to rule out explanations based on primitives such as probability weighting. Models such as non-cumulative or cumulative prospect theory (Kahneman and Tversky, 1979; Tversky and Kahneman, 1992) predict indifference in *Baseline Simultaneous*. Finding a strict preference for Asset U in *Baseline Simultaneous* thus provides evidence for bottom-up attention as a driver for overweighting unlikely events.

Fifth, since outcomes during the exposure stage of *Baseline Sequential* were not

²⁹Erev et al. (2008) argue that the description-experience choice gap is due to the typical presentation of lotteries in the description environment, e.g., in the form $((\$x, p; \$0, 1 - p) = (\$8, 0.1; \$0, 0.9))$. Such presentation makes likely and unlikely events appear equally prominent, which may equalize their perceived likelihood. When data from the same lottery is presented through sampling, the unlikely outcome is observed fewer times, affording a presentation that reflects the outcome’s actual likelihood. Our design accounts for this by presenting the outcome associated with each state graphically and with frequency proportional to its likelihood (uniform) across both conditions.

Figure 4: Mental Representations. Difference in perceived expected return of Asset F versus Asset U by information exposure condition.



payoff-relevant, this eliminates reinforcement learning as an explanation for our results (Erev and Barron, 2005).

Finally, our design differs from prior work on risky choice by including a graphical depiction of the distributions of outcomes. The visual representation allows us to manipulate salience exogenously, as we discuss further in Section 5.

4 Empirical Results

This section reports the results on representations (Section 4.1) and choices (Section 4.2). For brevity, we present results from the 11-state round here; results from the 51-state version are qualitatively and quantitatively similar and we present them in the Internet Appendix. There were no significant differences with respect to the order in which representations and choices were elicited and we pool our analysis across both orders. The Internet Appendix contains additional results and sample splits.

4.1 Representations

We first analyze how the mental representation varies with the information exposure condition by comparing the *Baseline Sequential* and *Baseline Simultaneous* treatments. All data discussed in this section are presented in Fig. 4, which plots the difference in perceived expected value for Asset F and Asset U, and Table 1, which reports the average perceived expected values and perceived probabilities of outcomes above or below the median by asset and treatment condition. Both exhibits also include results from the mechanism experiments that will be discussed in Section 5.

When exposed to information sequentially, participants perceived Asset F to have a higher expected value than Asset U. Specifically, they overestimated the expected

Table 1: Mental Representations. Average perceived expected value for each condition and asset (Panel A) and average perceived probability of outcomes below or above the median outcome \$1 for each condition and asset (Panel B).

Condition	Asset U	Asset F	Diff: F - U	t-test	N
Panel A: Perceived expected value of asset (objective = \$1.00)					
Simultaneous Information					
(SI1) Baseline	\$1.07	\$1.00	\$-0.07***	-2.83	381
(SI2) No Saliency Bias	\$1.06	\$1.09	\$0.03	0.81	100
Sequential Information					
(SQ1) Baseline	\$1.00	\$1.17	\$0.16***	7.34	404
(SQ2) No Recall Bias	\$1.08	\$1.08	\$0.00	0.06	100
(SQ3.1) Interference Asset F	\$0.99	\$1.38	\$0.38***	7.77	102
(SQ3.2) Interference Asset U	\$1.20	\$1.08	\$-0.12**	-1.97	99
Difference Across Exposure Methods					
(SQ1-SI1) Baseline: Sequential - Simultaneous			\$0.23***	7.11	785
(SQ2-SI2) No Recall - No Saliency Bias			\$-0.03	-0.66	200
Panel B: Perceived probability below/above median outcome (objective = 45.45%)					
Simultaneous Information					
(SI1) Baseline: $\Pr(x < 1)$	39.09%	44.27%	6.46%***	3.46	381
(SI1) Baseline: $\Pr(x > 1)$	42.63%	41.35%			
(SI2) No Saliency Bias: $\Pr(x < 1)$	41.71%	42.39%	3.26%	1.14	100
(SI2) No Saliency Bias: $\Pr(x > 1)$	44.16%	41.58%			
Sequential Information					
(SQ1) Baseline: $\Pr(x < 1)$	48.40%	38.24%	-25.87%***	-12.58	404
(SQ1) Baseline: $\Pr(x > 1)$	35.04%	50.76%			
(SQ2) No Recall Bias: $\Pr(x < 1)$	38.08%	40.55%	-0.68%	-0.20	100
(SQ2) No Recall Bias: $\Pr(x > 1)$	37.87%	41.02%			
(SQ3.1) Interference Asset F: $\Pr(x < 1)$	47.98%	37.38%	-39.47%***	-8.05	102
(SQ3.1) Interference Asset F: $\Pr(x > 1)$	28.83%	57.70%			
(SQ3.2) Interference Asset U: $\Pr(x < 1)$	34.88%	49.67%	13.66%**	2.54	99
(SQ3.2) Interference Asset U: $\Pr(x > 1)$	42.35%	43.48%			
Difference Across Exposure Methods					
(SQ1-SI1) Baseline: Sequential - Simultaneous			-32.34%***	-11.60	785
(SQ2-SI2) No Recall - No Saliency Bias			-3.94%	-0.87	200

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$; t-statistics in Panel B are from a difference-in-difference estimate between the below- and above-median probability assessments.

value of Asset F as \$1.17, which was significantly higher ($p < .01$) than their accurate estimate of \$1.00 for Asset U (see Panel A SQ1 in Table 1 and the first bar of Fig. 4).³⁰ Recall that the objective expected value for each asset was \$1. Similarly, participants underestimated (overestimated) the likelihood of outcomes below (above) the median of \$1 for Asset F, with the reverse for Asset U (see Panel B SQ1 in Table 1). The difference in excess probability assigned to outcomes above versus below the median for Asset F relative to Asset U was 25.87 percentage points, which is large and significant ($p < .01$). Importantly, by comparing these estimates to the known objective benchmarks we can conclude that reported beliefs reflect an incorrect mental representation in the direction predicted by Prediction 2.

In contrast, when exposed to information simultaneously, the distortion in perceived returns reversed direction: participants overestimated the expected value of Asset U as \$1.07, which was significantly higher ($p < .01$) than their accurate esti-

³⁰Unless otherwise noted, p -values correspond to the difference between assets.

mate of \$1.00 for Asset F (see Panel A SI1 in [Table 1](#) and the second bar of [Fig. 4](#)). Participants also assigned higher (lower) probability to outcomes above (below) the median for Asset U, with the reverse for Asset F. This led to a difference in excess probability assigned to outcomes above versus below the median for Asset F relative to Asset U of -6.46 ($p < .01$) (see Panel B SI1 in [Table 1](#)). These reported beliefs reflect an incorrect mental representation in the direction predicted by [Prediction 1](#).

The mental representations differ substantively across the two exposure conditions. The difference-in-difference estimate of the excess perceived expected value for Asset F in the sequential versus simultaneous condition is large and significant: at \$0.23 ($p < .01$), it represents nearly a quarter of the objective expected value of the assets (see Panel A (SQ1-SI1) in [Table 1](#)). Similarly, the excess probability mass assigned to outcomes above versus below the median for Asset F versus U in the sequential versus simultaneous treatment differs by 32.34 percentage points ($p < .01$). This shows that participants are sizeably and significantly more optimistic about Asset F in the sequential setting than in the simultaneous setting.

Note that because the order of eliciting beliefs and choices was randomized, these results are not driven by a desire to report beliefs that are consistent with one's choices.

4.2 Choices

Our key prediction is that the exposure method affects choice via its impact on the mental representation. In line with this prediction, we find that decisions largely follow the representational distortions documented in [Section 4.1](#), in that participants chose the asset perceived to have the higher expected value in each environment. When exposed to information sequentially, participants overwhelmingly chose Asset F over Asset U: 84.90% compared to 15.10% ($p < .01$; see SQ1 in [Table 2](#) and the first bar of [Fig. 5](#)).³¹ In contrast, when exposed to information simultaneously participants favored Asset U: 41.99% chose Asset F while 58.01% chose Asset U ($p < .01$; see SI1 in [Table 2](#) and the second bar of [Fig. 5](#)). Only 5% of choices corresponded to indifference; removing those choices from the analysis does not change the results either qualitatively or statistically.

Strikingly, despite facing the same decision and observing the same information in both conditions, the way in which participants were exposed to information shifted their choices by more than 40 percentage points ($p < .01$; see SQ1-SI1 in [Table 2](#)). This dramatic difference highlights the importance of understanding how cognition interacts with the decision environment to determine choice behavior.

4.3 Alternative Explanations

Keeping the objective distributions and information constant across conditions rules out many alternative potential mechanisms such as prospect theory or sampling error.

³¹Unless otherwise noted, choice p -values test the difference from 50%.

Figure 5: Choice. Percent of choices for Asset F by information exposure condition.

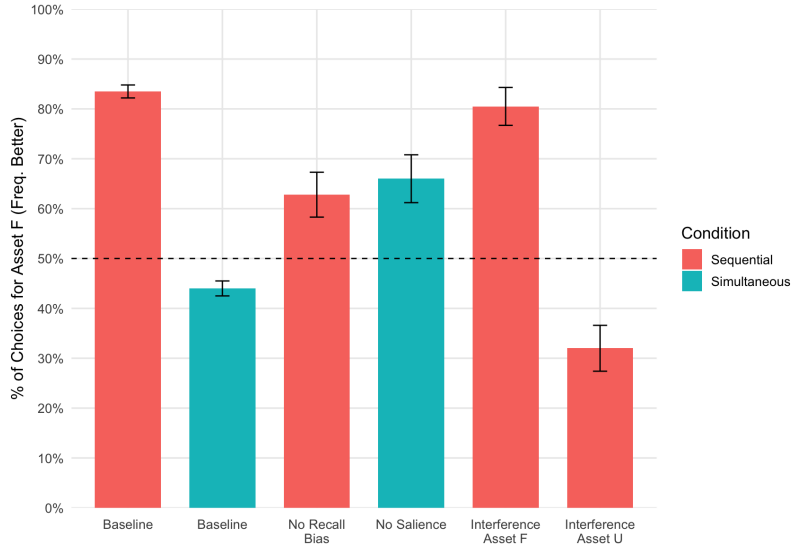


Table 2: Choice. Average share of participants selecting each asset.

Condition	Asset U	Asset F	Difference from 50%	<i>t</i> -test	N
Simultaneous Information					
(SI1) Baseline	58.01%	41.99%	-8.01%***	-3.16	381
(SI2) No Saliency Bias	34.00%	66.00%	16.00%***	3.36	100
Sequential Information					
(SQ1) Baseline	15.10%	84.90%	34.90%***	19.57	404
(SQ2) No Recall Bias	37.00%	63.00%	13.00%***	2.68	100
(SQ3.1) Interference Asset F	19.61%	80.39%	30.39%***	7.69	102
(SQ3.2) Interference Asset U	67.68%	32.32%	-17.68%***	-3.74	99
Difference Across Exposure Methods					
(SQ1-SI1) Baseline: Sequential - Simultaneous			42.91%***	13.98	785
(SQ2-SI2) No Recall - No Saliency Bias			-3.00%	-0.44	200

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

However, despite presenting participants with the same information in both conditions, it is possible that participants simply did not attend to the state where Asset U outperformed Asset F in *Baseline Sequential*. This would generate a gap in representations and choices through a mechanism similar to *sampling error* (Fox and Hadar, 2006) or *selective sampling* (de Palma et al., 2014). We analyze viewing times for each outcome pair as a measure of attention to explore this possibility. We find that participants pay significantly more attention to extreme outcome pairs—consistent with salience-driven attention. The average number of seconds participants take to view a moderate outcome pair is 1.87, whereas states with extreme differences receive 3.65 seconds of viewing time, or 1.78 seconds more attention on average ($p < .01$; see Table D.3). This near-doubling of attention for extreme outcome pairs confirms that the associated states are salient and not overlooked when participants are exposed

to information sequentially.

Yet as outlined in our framework, it is the set of outcomes that are available at the time of choice and not attention at the time of exposure that is key for driving decisions; if outcomes associated with a state are forgotten, they cannot be outweighed in the mental representation. Thus while salience does seem to drive attention to states with extreme differences at the time of exposure in *Baseline Sequential*, the bias generated by imperfect recall plays the dominant role in affecting representations and choices.

5 Exploring the Cognitive Mechanism

We next provide evidence for attention and memory as the drivers of the observed belief distortions by (i) measuring attention and (ii) exogenously manipulating factors associated with memory and attention constraints while keeping the underlying choice environment constant. The manipulations of attention and memory are motivated by the model in [Section 2](#). In particular, the model predicts that when information is presented simultaneously, manipulating salience away from state 11 will reduce the propensity to choose Asset U relative to Asset F. It also predicts that when information is presented sequentially, reducing memory constraints will reduce the propensity to choose Asset F and interference from other similar assets will impact choice.

5.1 Measuring Attention

We first test our assumption that the state with the largest contrast between payoffs relative to the local environment (state 11) attracts the most attention. We then show that channeled attention is predictive of choice.

We measure attention in three ways. First, we used the Mouselab paradigm of [Payne et al. \(1988\)](#).³² In this paradigm, participant’s attention to various objects in their choice environment is measured by the order in which they click on these objects, with the first-click interpreted as a proxy for the most attended to object.³³ We incorporated Mouselab by altering *Baseline Simultaneous* so that participants are required to click on a box under each state before proceeding to the next screen. Aside from this change, the *Mouselab Simultaneous Information* treatment was identical to *Baseline Simultaneous*, and was run on the same platform as the latter treatment. In line with our framework, if attention is directed towards state 11 and the focus on this state leads to a distortion in the representation that favors Asset U, then the data from this treatment will show: (i) participants are more likely to click on state 11 first relative to other states and (ii) participants who click on state 11 first will

³²The Mouselab design is a standard tool used in cognitive psychology to measure attention. It has been used to measure attention in consumer choice ([Reisen et al., 2008](#)), information acquisition ([Callaway et al., 2017](#)), and learning ([Ba et al., 2022](#)).

³³The assumption that the timing of click data corresponds to the ordering of attention has been validated using eye-tracking tools ([Meißner et al., 2010](#)).

Figure 6: Measuring attention. Percent of time each state clicked first in *Mouselab Simultaneous Information*.

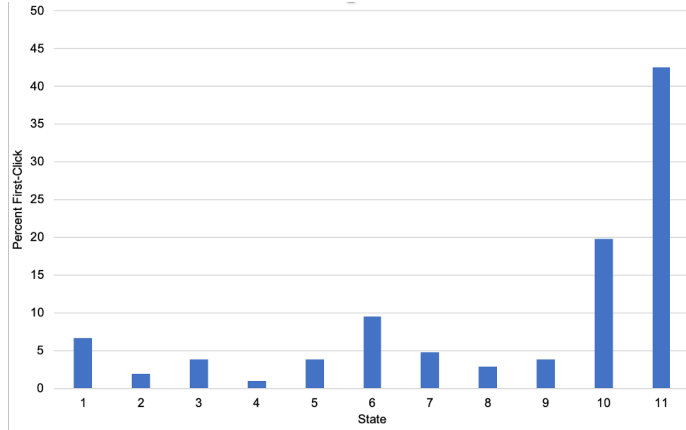


Table 3: Attention and Choice. Participants who clicked on the most salient state (state 11) first were significantly more likely to choose Asset U.

Choice of Asset U		
	Coefficient (t-statistic)	Standard Error [95% Conf. Interval]
First-Click	0.31***	0.09
State 11	(3.34)	[0.13, 0.49]
Constant	0.45***	0.07
	(6.18)	[0.31, 0.59]
Number of observations	106	

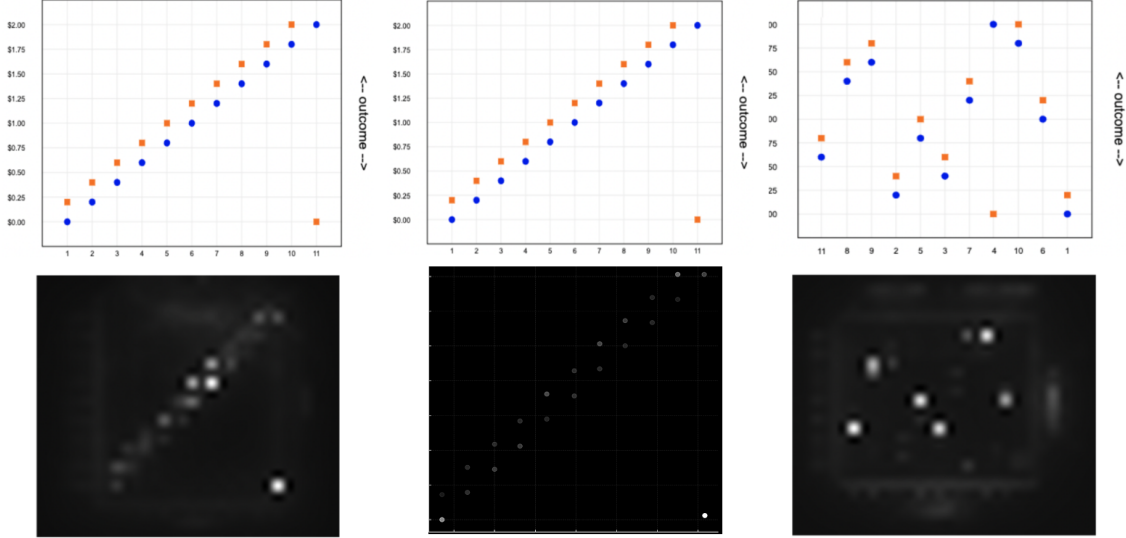
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

be more likely to choose Asset U over Asset F.

Data on first click by state is presented in Fig. 6. Indeed, attention is channeled towards state 11: over 40 percent of participants clicked on state 11 first, which is more than twice the share that clicked on the second-most-clicked-first state (state 10), and roughly eight times the share that clicked on any of the other states first. Turning to choice, Table 3 regresses choice of Asset U on attending to state 11 the most—as proxied by clicking on it first. Indeed, participants who attended to state 11 the most were more than 30 percentage points more likely to choose Asset U. Together, these findings support our assumption that state 11 is most salient and directing attention towards state 11 increases the propensity to choose the higher performing asset in that state, i.e., Asset U.

Second, we measured attention using the webcam-based eye-tracking method developed by Yang and Krajbich (2021). Participants began the study by providing normalization data to standardize their gaze within our field of interest. Participants then proceeded as in the *Baseline Simultaneous* treatment. We measured their at-

Figure 7: Measuring Saliency. Saliency of each state in (i) *Baseline Simultaneous* using Graph-Based Visual Saliency model (left) (ii) *Baseline Simultaneous* using Eye-tracking (middle) and (iii) *No Saliency Bias* using Graph-Based Visual Saliency model (right). Top image depicts distributions shown to participants and bottom image depicts saliency maps generated by corresponding tool.



tention as a function of dwell time on each of the potential outcomes of the assets when they were presented with the distribution of returns.³⁴ The middle column of Fig. 7 presents the data on dwell time, with brighter dots indicating more dwell time. As can be seen in the figure, this measure was greatest for the Asset F low outcome in state 11 (bottom right): the attention weight on outcomes associated with this state is roughly 1.5 times larger than that the weight on the outcomes associated with the second-most-attended-to state, and more than twice the average attention weight across all states.

Finally, we used the bottom-up Graph-Based Visual Saliency model and the associated algorithm proposed in Harel et al. (2006) to predict what state in *Baseline Simultaneous* would draw the most attention.³⁵ The left column of Fig. 7 presents the output of the model. As predicted, the low outcome for Asset F in state 11 is the most salient: it had an average activation level of approximately 230 pixels while

³⁴This measure is motivated by a literature which has shown that higher dwell time in a region of interest is predictive of choice (Mullett and Stewart, 2016; Orquin and Mueller Loose, 2013). We computed a measure of attention for each outcome in the graph as follows: we assigned each gaze point to the nearest point describing an asset’s outcome, with the weight increasing as a function of dwell time, and summed these weighted points to obtain an attention measure for each outcome.

³⁵In short, the model generates a saliency map of an image using an algorithm that was validated through measures of human visual fixation. The output of the saliency map is based on levels of activation in units of pixels—higher pixels/brighter areas are associated with higher levels of saliency. Bose et al. (2022) have recently used this technique to show how saliency can predict choice between financial assets.

the next highest outcome had an activation level of less than 100.

The Mouselab and eye-tracking methods are complementary. While the Mouselab paradigm uses a proxy for attention rather than a closer cognitive analogue such as gaze in the eye-tracking paradigm, having a single data point (first-click) allows us to correlate channeled attention with choice. Moreover, the results from the Mouselab variation help validate the Graph-Based Saliency model approach in our setting: the two align in terms of where attention is directed and predicted to be directed.

5.2 Manipulating Saliency

We next exogenously manipulate saliency without changing the distribution of the assets. By keeping the distribution fixed, we can directly test the impact of distorted mental representations on choice while keeping underlying preferences constant.³⁶ We manipulated saliency by reordering the states from the predictably-increasing pattern in presented in the *Baseline Simultaneous* treatment. In that treatment, aside from state 11, differences in outcomes in adjacent states are small and constitute small deviations from the local environment—thus making the large difference in state 11 all the more salient. In contrast, under the reordering in the new *No Saliency Bias* treatment, adjacent states could have small, medium, or large differences in outcomes. Thus, the large outcome difference in state 11 represents a smaller deviation from the local environment and therefore is less salient. The return chart used in *No Saliency Bias* is depicted in the top right image of Fig. 7. Aside from the different return chart, the treatment was identical to *Baseline Simultaneous*.

We first used the Graph-Based Visual Saliency model and algorithm to test whether state 11 was still most salient in *No Saliency Bias*. As shown in the bottom right image of Fig. 7, no one state is predicted to be dominant in terms of saliency. In contrast with *Baseline Simultaneous* where state 11 had the highest activation level, state 11 in *No Saliency Bias* had an average activation level of approximately 30 pixels while six other states had activations levels in the range of 100 to 150 pixels. Thus while channeled attention may still distort the mental representation, it will not do so systematically in favor of Asset U.

Turning to the data on mental representations and choice, the mental representation no longer featured a systematic distortion towards Asset U (see the fourth bar of Fig. 4 and treatment SI2 in Table 1): there was no significant difference in perceived expected value between the two assets. Following this pattern of beliefs, choices shifted significantly from Asset U to Asset F (see the fourth bar of Fig. 5 and treatment SI2 in Table 2). The elimination of both the systematic bias in perceived expected return favoring Asset U and the propensity to choose Asset U suggests that saliency-channeled attention plays an integral role in mental representations and de-

³⁶Prior work in the domain of risky choice manipulated saliency by manipulating payoffs (Bordalo et al., 2013). This approach may confound differences in representation with differences in preferences.

cisions when people are exposed to a large set of information at the same time. By holding fixed the payoff structure in *Baseline Simultaneous* and *No Salience Bias*, the difference in behavior between these two treatments also provides further evidence that deviations from the local environment are a key driver of salience—and not just correlation between alternatives (Leland, 1998). This rules out correlation-sensitive models such as regret theory (Loomes and Sugden, 1982) in explaining our results.

5.3 Manipulating Memory

We now proceed to exogenously manipulate factors related to memory in the sequential information setting. Specifically, we run treatments that reduce recall bias by providing a memory aide, introduce interference from an additional decoy asset, and reduce the likelihood of forgetting the most salient state by manipulating the recency property of memory.

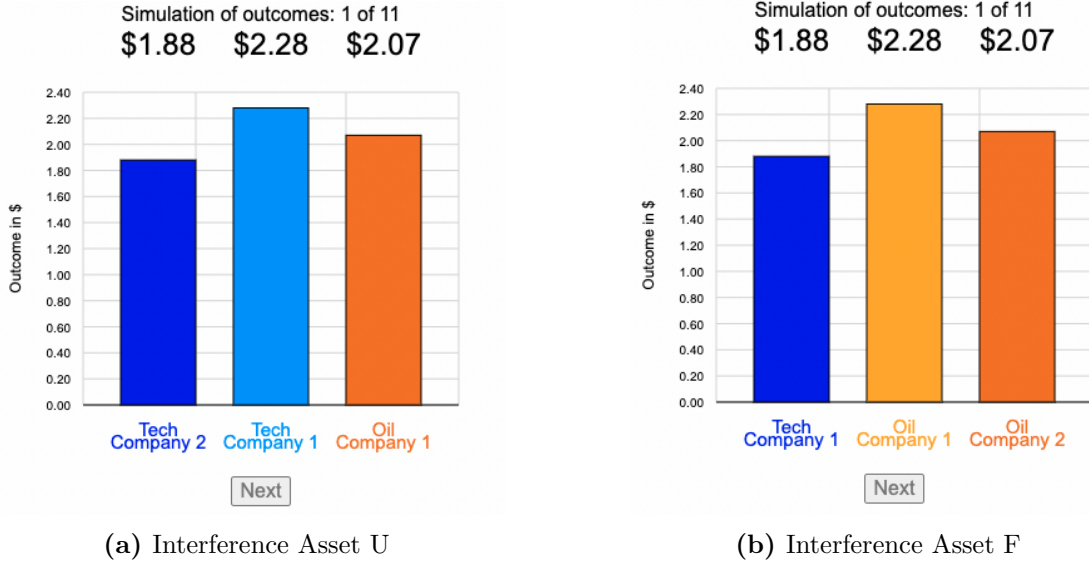
Reducing Recall Bias. First, we sought to relax the constraint on memory. In *No Recall Bias*, participants were asked to write down outcomes during the sequential sampling process. Specifically, they were asked to have a pen and paper (or other method of recording) ready and incentivized with a potential bonus payment for correctly responding to a quiz at the end of the round. Aside from these instructions, the treatment was identical to *Baseline Sequential*. By reducing the scope for imperfect recall, participants in *No Recall Bias* are predicted to exhibit a smaller representational distortion towards Asset F compared to *Baseline Sequential*, and therefore, a lower propensity to choose Asset F.

Mental representations and choices are in line with these predictions. The mental representation no longer featured a systematic distortion towards Asset U (see the third bar of Fig. 4 and treatment SQ2 in Table 1): in contrast to *Baseline Sequential*, there was no significant difference in perceived expected value between the two assets. Following this pattern in the elicited representations, choices shifted significantly from Asset F to Asset U (see the third bar of Fig. 5 and treatment SQ2 in Table 2): relative to *Baseline Sequential*, the share of choices for Asset F decreased by roughly 20 percentage points ($p < .01$).

Notably, as can be seen from comparing the third and fourth bars in Figs. 4 and 5, *No Salience Bias* and *No Recall Bias* eliminated the belief and choice gaps between simultaneous and sequential information exposure: participants held statistically indistinguishable beliefs and chose each asset with similar propensity regardless of whether information was displayed simultaneously or sequentially (see SQ2-SI2 in Tables 1 and 2). This demonstrates the role of constrained attention and memory as drivers of the gap between simultaneous and sequential exposure in the baseline conditions, rather than a feature inherent to the exposure method per se.

Interference. Next, we exploited the interference property of associative memory by introducing a decoy asset (Asset D) which was either similar to Asset U or Asset

Figure 8: Information display in interference treatments. Asset D (middle bar) dominated both Asset U (left bar) and Asset F (right bar).



F. Participants proceeded through a task that was otherwise identical to the *Baseline Sequential* treatment, aside from also being presented with the outcomes for Asset D. They did not have the option to choose it: each sequential sample included the outcomes for Assets F, U, and D but the choice remained between Assets F and U.³⁷ The decoy asset’s outcomes dominated both Asset F and U, so that it was more attractive than either alternative. We introduced payoff-independent features to the assets in order to exogenously manipulate the similarity of Asset D to the other two assets. As shown in Fig. 8, we used color (shades of orange vs. blue) and industry classification (oil vs. tech) to structure the decoy Asset D to be either similar to Asset U (*Interference Asset U*) or to Asset F (*Interference Asset F*). Participants were randomized into one of these two conditions.

This design sought to test [Prediction 3](#), which asserts that the decoy will inflate the perceived expected value of the asset similar to it. Participants in *Interference Asset F* are predicted to continue perceiving Asset F as having a higher expected return, and therefore, to continue choosing Asset F more frequently than Asset U. In contrast, participants in *Interference Asset U* are predicted to exhibit a smaller representational distortion in favor of Asset F—or potentially even a reversal to favor Asset U—and therefore, a lower propensity to choose Asset F than in *Baseline Sequential*.

³⁷Participants were informed that their choice was between Assets F and U before observing any information about the assets. As in the baseline, Assets F and U had the same marginal distribution. These marginal distributions are slightly different from *Baseline Sequential*: outcomes are on an irregular grid to have more digits, which is necessary to avoid systematic differences relative to the decoy in the “roundness” of outcomes. Asset F still outperforms Asset U by a moderate amount in 10 states and underperforms by a large margin in state 11. See [Table D.2](#) for the joint distribution.

In line with this prediction, in *Interference Asset U* participants perceived Asset U to have a higher expected value than Asset F (see the sixth bar of Fig. 4 and treatment SQ3.2 in Table 1). The distortion in the elicited representation reverses compared to *Baseline Sequential*: the gap in perceived expected value flipped from \$0.16 favoring Asset F to \$0.12 favoring Asset U ($p < .01$). In contrast, in *Interference Asset F* participants perceived Asset F to have a higher expected value than Asset U (see the fifth bar of Fig. 4 and treatment SQ3.1 in Table 1). Compared to *Baseline Sequential*, representations were distorted even more in favor of Asset F: the gap in perceived expected value increased from \$0.16 to \$0.38 favoring Asset F.

Choices followed the pattern in representations. When the decoy asset was more similar to Asset F, the propensity to choose Asset F was reinforced: participants selected Asset F at around the same rate in *Interference Asset F* and *Baseline Sequential* (see the fifth bar of Fig. 5 and treatment SQ3.1 in Table 2).³⁸ However, this propensity to choose Asset F reversed when the decoy asset was more similar to Asset U: 63% of participants chose Asset U in *Interference Asset U*, compared to 15% in *Baseline Sequential* (see the sixth bar of Fig. 5 and treatment SQ3.2 in Table 2).³⁹ Comparing the two interference conditions, whether the decoy asset was similar to Asset U versus Asset F shifted choices by nearly 50 percentage points. Notably, these results highlight how cognitive factors—here, similarity-driven interference—can drive choices towards the asset with rare out-performance even in the sequential information environment.

Recency Bias. To provide further evidence for the imperfect recall mechanism, we leveraged the recency property of memory: more recent events are more likely to be remembered than those further in the past. Specifically, we modified *Baseline Sequential* so that state 11—where Asset U has a higher outcome than Asset F—was presented last in the sequence. We also included a question immediately after the sequence which asked participants which of the two assets had a higher outcome in the most-recently-viewed state as a manipulation check. The recency property of memory predicts that displaying state 11 last will increase the probability it is remembered correctly. In turn, this *Recency* condition is predicted to decrease the difference in perceived expected value between Asset F and Asset U and decrease the propensity to choose Asset F over Asset U compared to *Baseline Sequential*.

The majority of participants in the *Recency* condition remembered the state where Asset U had a higher outcome (at least directionally), with 77% answering the manipulation check correctly. Moreover, the belief distortion in favor of Asset F decreased

³⁸Even though the mental representation was distorted even further in favor of Asset F in this condition, the already very high prevalence of choosing Asset F in *Baseline Sequential* likely led to a ceiling effect.

³⁹A simple dislike for a specific color or industry cannot explain this result, as this mechanism would generate a similar change in the fraction of choices for Asset F across both interference conditions.

substantially: the 0.06 difference in perceived expected value in *Recency* was significantly smaller than the same statistic in *Baseline Sequential* ($p < .01$) and no longer significantly different from zero. Choices followed suit: in *Recency*, participants were nearly 20 percentage points less likely to choose Asset F relative to *Baseline Sequential* ($p < .01$).

5.4 Discussion

Eliminating the gap in mental representations between the two assets allows for isolation of any underlying preference for either asset. In *No Salience Bias* and *No Recall Bias*, participants reported the same perceived expected value for Assets F and U (see Fig. 4) but chose Asset F about 20 percentage points more often than Asset U (see Fig. 5).⁴⁰ Given that beliefs are the same for both assets, the higher propensity to choose Asset F reveals a preference for it. This highlights an important implication of our framework: by eliciting the mental representation we are able to attribute remaining choice differences to preferences.

While it is beyond the scope of the current paper to determine why preferences favor Asset F, one may consider two possibilities. First, a rich literature in psychology and economics has highlighted the role of noisy cognition in evaluating prospects (see Woodford (2020) for review). Cognitive noise in the evaluation stage leads to attenuation and an insensitivity in the behavioral response to changes in the fundamentals of the choice problem (Enke and Graeber, 2023; de Clippel et al., 2024). In our setting, cognitive noise attenuates the large outcome difference favoring Asset U in state 11 relative to the smaller frequent differences favoring Asset F in the other ten states. Thus absent representational distortions, cognitive noise would push choices towards Asset F. Second, participants may have a basic preference for alternatives that offer small wins while avoiding large potential losses. Robson (1996) shows that this type of preference can emerge as a result of fitness maximization.

6 Applications

We next show how our framework can explain previously-documented choice anomalies and be generalized beyond risky choice.

6.1 Constrained Attention and the Four-Fold Pattern of Risk

Our results suggest that anomalies that appear to over- or underweigh rare events may be driven at least in part by the way constrained attention distorts the mental representation. For instance, the four-fold pattern of risk corresponds to the documented pattern where people are (i) risk-seeking over low-probability gains and high-probability losses and (ii) risk averse over high-probability gains and low-probability losses. This pattern has typically been attributed to preferences—specifically probability weighting, which corresponds to the tendency to overweigh (underweigh) low

⁴⁰Note that the propensity to choose Asset F in these conditions is substantially lower than in *Baseline Sequential*, where the mental representation was distorted in favor of Asset F.

(high) probability outcomes (Tversky and Kahneman, 1992).

We use a variation of our simultaneous information paradigm, where an asset has an unlikely salient low or high outcome, to test whether salience-channeled attention can generate the four-fold pattern of risk. Our framework predicts that participants will overweigh the unlikely salient state, leading them to underestimate the asset’s return and exhibit risk aversion in the case of the low outcome, and to overestimate the return and exhibit risk seeking in the case of the high outcome.

The design was as follows. Participants faced a series of decisions between two assets, A and B. The outcome distribution of Asset A remained the same across all decisions. In the *Salient Low Outcome (G90)* condition, it assigned an outcome of \$20 to 9 out of 10 states and \$0 to the final state. Thus it had a high probability of a high payoff (\$20) and a low probability of a low payoff (\$0). In the *Salient High Outcome (G10)* condition, it assigned an outcome of \$0 to 9 out of 10 states and \$20 to the final state. Thus it had a high probability of a low payoff and a low probability of a high payoff. As before, the distribution over states was uniform.

The outcome distribution of Asset B changed across decisions. The outcome was the same for all states, but this outcome increased by increments of \$1 from \$0 in the first decision to \$15 in the final decision. The idea was that a participant would initially prefer Asset A when Asset B’s (certain) payoff was low and switch to preferring Asset B at some point as its payoff increased. Her indifference point corresponded to the point at which the switch occurred, and the value of Asset B at this switch point provided an estimate of her certainty equivalent for Asset A. Comparing this certainty equivalent to the expected value of Asset A determined whether the participant exhibited risk aversion or risk seeking behavior.

Panel A of Fig. 9 illustrates the outcome distributions in one of the *Salient Low Outcome (G90)* decisions, where Asset B has a certain payoff of \$18. State 10 features the largest deviation in outcomes relative to the local environment and is therefore the most salient.⁴¹ By the same logic as in *Baseline Simultaneous*, our framework predicts that focusing on state 10 where Asset A returns less than Asset B will lead to a distorted mental representation that underestimates the return for Asset A, leading to a perceived expected return below its expected value and choices consistent with risk aversion. Thus, *Salient Low Outcome (G90)* is predicted to generate choices that match part (ii) of the four-fold pattern, i.e., risk aversion over the high-probability gain of \$20 in states 1 through 9. The intuition for *Salient High Outcome (G10)* is analogous: our framework predicts that participants will overestimate the return for Asset A, leading to a perceived expected return below its expected value and choices consistent with risk seeking. Such choices match parts (i) of the four-fold pattern, i.e., risk seeking over the low-probability gain of \$20 in state 10.

⁴¹We used the Graph-Based Visual Saliency model to validate this assumption. Indeed, the method predicts that state 10 is most salient.

Figure 9: Outcome distributions for four-fold pattern of risk conditions.

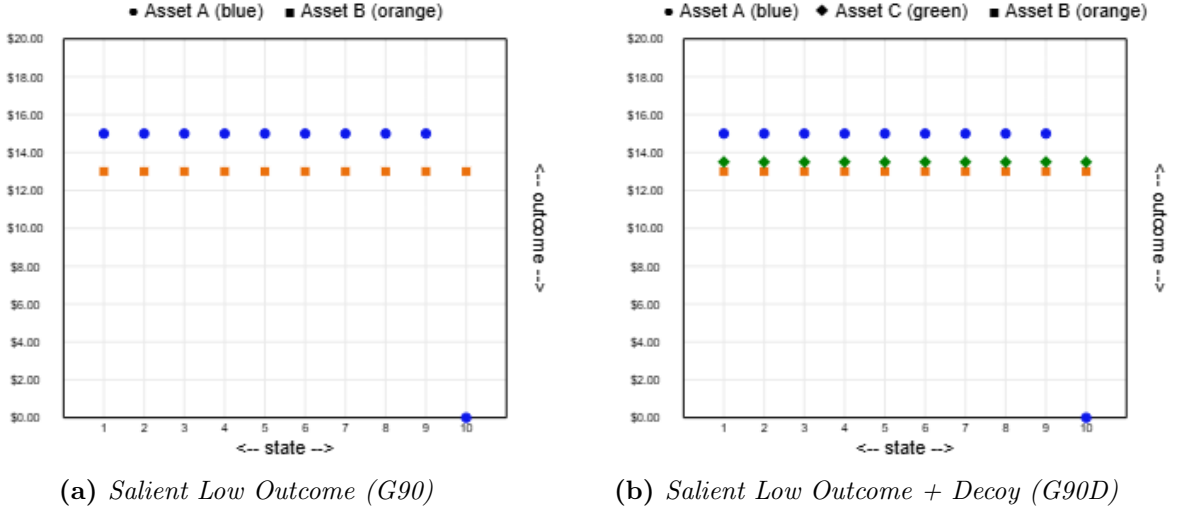


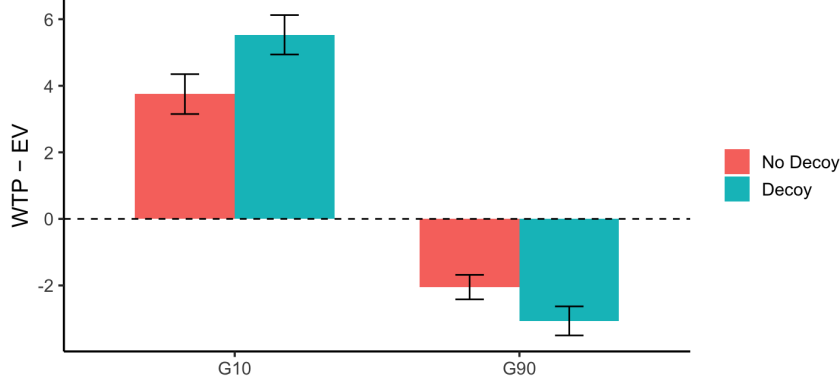
Fig. 10 presents results from both conditions. In *Salient High Outcome* where there was a low probability high outcome, the certainty equivalent of Asset A exceeded its expected value, which is consistent with risk seeking (left pink bar in figure). In contrast, in *Salient Low Outcome* where there was a low probability high outcome, the certainty equivalent of Asset A was less than its expected value, consistent with risk seeking (right pink bar in figure).

Manipulating Salience. To provide direct evidence for the attentional mechanism, we ran variations of *Salient Low Outcome* and *Salient High Outcome* that featured a decoy asset (Asset C) that returned the expected value of Asset A in all states. Panel B of Fig. 9 illustrates the returns chart in the *Salient Low Outcome + Decoy* condition. Note that, as in the previous decoy treatments, Asset C was not available for choice. The presence of the decoy asset concentrated more outcomes around the expected value of Asset A, which makes the contrast in outcomes in state 10 even more salient.

If the four-fold pattern is driven by constrained attention, then increasing the salience of state 10 will exacerbate this pattern. Indeed this is exactly what we find: the difference between the certainty equivalent and expected value of Asset A is larger in both decoy conditions (see green bars of Fig. 10 and Table D.15). Thus participants appear to exhibit a higher level of risk seeking in *Salient High Outcome + Decoy* compared to *Salient High Outcome*, and a higher level of risk aversion in *Salient Low Outcome + Decoy* compared to *Salient Low Outcome*. Since the choice options and valuation method were held constant across the conditions with and without a decoy, these results provide direct evidence for salience-channeled attention as a driver of the four-fold pattern in our setting.

This finding is notable because the four-fold pattern is a choice anomaly generally attributed to preference primitives (Tversky and Kahneman, 1992). Oprea (2024)

Figure 10: Four-fold pattern of risk. Participants exhibited risk seeking over low probability gains (*Salient High Outcome (G10)*) and risk aversion over high probability gains (*Salient Low Outcome (G90)*). Bars depict difference between certainty equivalent and expected value of Asset A.



argues that the four-fold pattern is generated by responses to complexity. Our results highlight the role of a specific form of complexity (representational) and identify constrained attention as the psychological mechanism.⁴²

6.2 Non-Risky Choice

It is straightforward to apply our framework to non-risky choice, where constrained attention and memory impact the representation of complex choice environments without risk. We test this hypothesis using variations of *Baseline Sequential* and *Baseline Simultaneous*.

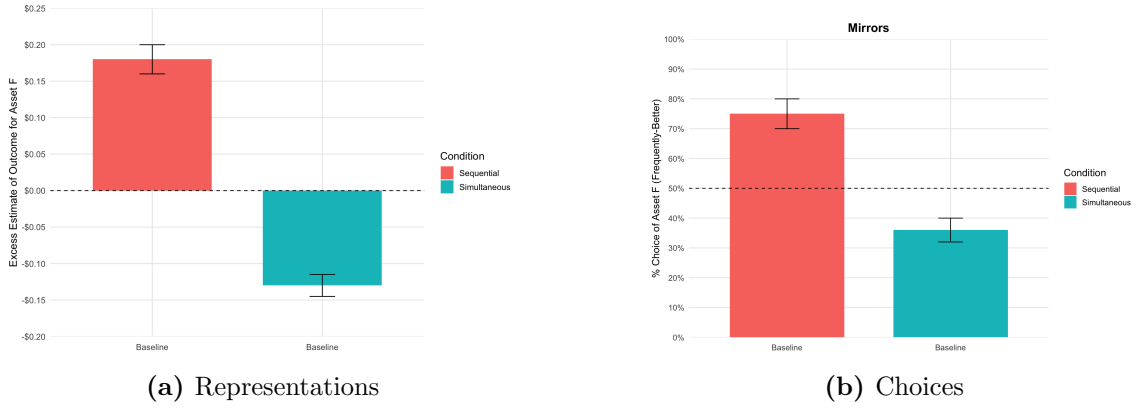
Following Oprea (2024), who created non-risky analogues of risky choice environments by paying participants their average payoffs across all states (as opposed to their payoff in a randomly drawn state), we created non-risky analogues of *Baseline Sequential* and *Baseline Simultaneous*. Specifically, rather than being paid the outcome of their chosen asset in a randomly-drawn state, participants were paid the *average* outcome of their chosen asset with certainty.⁴³ Note that the average outcome corresponded to the expected value of the asset; thus, both Assets U and F had the same certain payment of \$1. We elicited their representations as the perceived payment for each asset (rather than average payment, as in the baseline risky settings). Aside from this change, the *Mirrors Sequential* and *Mirrors Simultaneous* treatments were identical to the corresponding baseline treatments.

Similar to the risky choice baseline conditions, participants overestimated the

⁴²Oprea (2024) provides evidence for complexity as the driver by replicating the same choice pattern in an environment without risk. We ran deterministic version of the four conditions in Section 6.1. Our results were nearly identical to the pattern depicted in Fig. 10. See Fig. C.4 for the results.

⁴³Participants were told that payments across all states (“situations”) would be added up and divided by the number of states.

Figure 11: Non-Risky Choice. Difference in perceived expected payoff for Asset F versus Asset U (Panel A) and percent of choices for Asset F (Panel B) by information exposure condition (*Mirrors Sequential* or *Mirrors Simultaneous*).



payoff associated with Asset F when exposed to information sequentially and overestimated the payoff associated with Asset U when exposed to information simultaneously (both $p < .01$; see Panel A of Fig. 11). In both cases, the perceived expected values of the two assets different significantly from each other. Choices reflected this distortion in perceived payoffs: participants were more likely to choose Asset F in *Mirrors Sequential* and Asset U in *Mirrors Simultaneous* (both $p < .01$ relative to 50%; see Panel B of Fig. 11). These results suggest that our constrained attention and memory framework is also applicable to non-risky choice settings.

7 Related Literature

Experience effects. Our results contribute to the literature on experience effects in consumer choice and financial decision making. Studies have documented long-lasting effects of personal experiences on economic decision-making; for example, [Malmendier and Nagel \(2011\)](#) find that negative individual stock-market experiences lead to more conservative investing decades later (see [Malmendier and Wachter \(2021\)](#) for review). This literature examines the influence of an individual’s personal sample compared to the full historical sample, finding that the former is overweighted in both beliefs and decisions.⁴⁴ We show that people can appear to either over- or underweight unlikely ‘extreme’ states depending on the exposure environment. Our findings suggest that while ‘extreme’ events are underweighted when exposed to sequential information alone (e.g., experiencing observations throughout one’s life), factors that are important for memory recall such as cue-based similarity (e.g., media coverage of similar events) or interference can lead to them to appear to be overweighted in decisions. Another strand of literature shows that generating *artificial* experience through sequential sampling can be used to affect beliefs about returns

⁴⁴The personal experience literature does not examine the weighting of unlikely “extreme” states compared to more likely “moderate” states within the individual’s sample, which is the focus of our study.

and investment choices (e.g., [Kaufmann et al., 2013](#); [Laudenbach et al., 2022](#)). We contribute to this research on natural and artificial experience effects by highlighting the interaction between bounded cognition and how people encode information as the essential link between beliefs and choices in sequential exposure environments.

Description-Experience gap. Our findings have implications for the observed “decision-experience” gap in risky choice (see [Hertwig et al. \(2004\)](#), and [de Palma et al. \(2014\)](#) for review). In prior work documenting this gap, people select between a safe option and a risky gamble that has a large but unlikely payout. In the description environment, information about the outcomes and their respective likelihoods is presented simultaneously. In the experience environment, people sequentially sample from the distribution of the gamble, with replacement; they can stop sampling whenever they want and then proceed to make a choice. People were more likely to select the risky asset in the former environment than the latter. Several explanations have been proposed for this shift in choices. In the case of the description condition, the higher preference for the risky asset has been attributed to probability weighting, which leads to the overweighting of unlikely events ([Kahneman and Tversky, 1979](#)). In the model, objective probabilities are transformed through a probability-weighting function—typically assumed to be a preference primitive—which leads people to overweight low probability outcomes and underweight high probability outcomes ([Tversky and Kahneman, 1992](#)). Some have argued that the apparent underweighting of unlikely events when exposed to information through experience is due to differences in the information set at the time of choice. For example, [Fox and Hadar \(2006\)](#) propose that people are more likely to go for the safe option because they do not even know about the unlikely payout of the gamble—they simply may not sample enough to have the same information about the distribution as those in the experience treatment (see [Cubitt et al. \(2022\)](#) for a similar conjecture). Others argue that the gap is driven by reinforcement learning ([Erev and Barron, 2005](#)) or simple tallying heuristics ([Hills and Hertwig, 2010](#)). Notably, [Erev et al. \(2010\)](#) do propose limited memory as one of many potential explanations, but do not test this mechanism directly.

We derive the “description-experience” gap from first principles—namely, bounded attention and memory—and present direct evidence for these mechanisms within the same paradigm. Importantly, we predict and show that the gap in choices is driven by a difference in the basic *representations* of the choice environment: people’s choices reflect differences in what problems they think they’re facing.⁴⁵ We do so in a setting

⁴⁵Contemporaneous work by [Oprea and Vieider \(2024\)](#) offer a similar and complementary explanation for the “description-experience” gap. Their explanation is based on a combination of noisy coding and biased sampling, which generates the gap in two-state settings such as the ones examined in prior work on the topic. As we highlight in [Section 2](#), limited attention and memory are not predicted to bite in such simple settings; these constraints are relevant as representational complexity increases, which is the focus of our study. Our experimental setting also rules out the

that is explicitly designed to rule out the explanations that have been previously proposed in the literature. The experiment 1) involves a choice between assets with the same outcome distribution (which rules out primitives such as probability weighting), 2) ensures that people have the same information about the distribution across decision environments (which rules out sampling error), 3) abstracts from differences in hedonic feedback before the choice is made (which rules out reinforcement learning), and 4) allows for the explicit manipulation of factors affecting attention and memory (which allows for identification of the mechanism). The fact that we can eliminate and even reverse the “description-experience” gap by manipulating the relevant cognitive constraints, as well as demonstrate that gap in non-risky choice, suggests that the gap is not driven by something inherent to learning from description or experience per se. Our results also show that preference-based mechanisms for the gap, such as a preference for frequent “wins” (Olschewski et al., 2024), are unlikely to rationalize it.

Cognitive constraints and behavioral anomalies. Our findings inform the line of work exploring how cognitive constraints reconcile and rationalize seemingly anomalous economic phenomena in a variety of domains. Models of bottom-up attention and salience are used to explain the Allais paradox and risk-seeking behavior (Bordalo et al., 2012), insensitivity to taxes (Chetty et al., 2009; Taubinsky and Rees-Jones, 2018), and behavior in strategic interactions (Li and Camerer, 2022); see Bordalo et al. (2022b) for a review. Models of bounded memory can explain projection bias, context-dependent consumer choice, availability and representativeness heuristics (Bordalo et al., 2022a, 2020), as well as overreaction to information (Enke et al., 2020; Hartzmark et al., 2021), and belief-decay (Graeber et al., 2023). Recent studies build on evidence from neuroscience to model how the difficulty of perceiving and adapting to outliers can explain the neglect of outlier risk (d’Acremont and Bossaerts, 2016; Payzan-LeNestour and Woodford, 2022). Models of constructed preferences are used to rationalize anchoring effect and provide interpretations for seemingly-coherent demand curves (Ariely et al., 2003). Finally, models of cognitive noise and complexity are used to explain small-stakes risk aversion (Khaw et al., 2021), insensitivity to decision-relevant parameters (Enke and Graeber, 2023), the four-fold pattern of risk (Oprea, 2024), and non-constant time discounting (Enke et al., 2023). We contribute to this literature by identifying an important interaction between cognitive constraints and the information exposure method, and documenting its effect on mental representations of information and choice behavior.

Our paper is most related conceptually to how cognitive constraints distort people’s representations of their choice or informational environment. Steiner and Stewart (2016) theoretically argue that cognitive noise in information processing leads to a winner’s curse effect in choice—namely, unbiased noise in perceiving the value

noisy cognition and biased sampling mechanisms by design.

of an option leads people to overvalue the option that they end up choosing. They argue that perceiving low (high) probability events as more (less) likely is an optimal heuristic developed to account for this winner’s curse in choice. [Bordalo et al. \(2025\)](#) theoretically show that constraints on memory and attention can lead people to neglect features of the information environment, which leads them to use exact heuristics such as reporting the prior or signal diagnosticity in a belief-updating task; [Fan et al. \(2025\)](#) provide evidence for the use of such exact heuristics in a lab experiment and [Bastianello et al. \(2025\)](#) show that this mechanism generates ‘sparse’ mental representations in a sample of professional forecasters. [Jiang et al. \(2025\)](#) show that recalled experiences have greater explanatory power in explaining investor expectations than their actual experiences. Finally, [Ba et al. \(2022\)](#) show that constraints on attention and working memory lead people to focus on a limited number of salient states when processing new information; this leads them to overreact to information as the representational complexity in the information environment increases—operationalized as the number of states—and underreact when the information environment is simple.

Frictions in financial markets. We also contribute to the more conventional portfolio choice and asset pricing literature by varying dependence while carefully holding marginal distributions constant. Dependence is intrinsic to models of optimal portfolio choice and capital market equilibrium going back to [Markowitz \(1952\)](#) and [Sharpe \(1964\)](#). In these models, dependence between asset returns is important because it results in non-diversifiable risk (systematic risk or ‘beta’), which makes portfolios more risky for individuals and depresses prices in the aggregate. Going beyond the effect of dependence on portfolio risk, we provide the underpinnings of recent studies focusing on effects of dependence on beliefs, portfolio choice, and asset prices when investors learn from realistic samples of returns and prices ([Ungeheuer and Weber, 2021, 2023](#)). Incorporating dependence is unusual in behavioral finance, where typical approaches assume narrow framing when investors evaluate individual assets (e.g. [Barberis et al., 2016](#), where investors evaluate stocks in isolation using the Cumulative Prospect Theory value function and a sample of 60 monthly returns). The few exceptions that explicitly consider dependence being [Cosemans and Frehen \(2021\)](#), who correlate historical stock returns with return predictors based on Salience Theory, and [Barberis and Jin \(2023\)](#) who consider the implications of reinforcement learning for investor behavior. We contribute to this literature by showing how dependence can affect capital markets through a fundamentally new interaction between how people are exposed to information, memory and attention mechanisms (not related to portfolio risk), and the formation of people’s mental representations.

8 Discussion

We explore how cognitive constraints lead to distortions in people’s mental representations of their choice environments. We propose a simple framework where attention and memory constraints interact with how people are exposed to information. The framework generates predictions on how simultaneous versus sequential information exposure leads to distinct and systematic representational distortions. Experimentally eliciting representations and choices allows us to test these predictions and isolate the role of biased representations in choice behavior. These findings have several implications for our understanding of risky choice.

First, understanding the role of biased representations in choice behavior is important for identifying preferences, as the potential for distorted mental representations generates an identification problem where multiple preferences and representations are consistent with observed choice data.⁴⁶ If the information setting systematically distorts mental representations, then one needs both choice and representational data in order to identify preferences and conduct welfare analysis. This insight has implications for understanding the origin of documented behavioral anomalies, including those previously attributed to preferences.

Bounded attention and memory provide one explanation for heterogeneity in mental representations and risky choice (e.g., [Diecidue et al., 2020](#)). In a given environment, participants may differentially misremember unlikely states or differentially have their attention drawn to salient states, leading to heterogeneity in choice. Moreover, differences in information exposure settings, where some people learn from simultaneous information while others learn sequentially, will generate predictable heterogeneity in risk perceptions, beliefs, and behavior. For example, research in finance has documented that investors are attracted to lottery-like stocks and out-of-the-money options that outperform infrequently but mostly underperform (e.g., [Barberis and Huang, 2008](#); [Kumar, 2009](#); [Boyer and Vorkink, 2014](#); [Bryzgalova et al., 2023](#)). They are also attracted to prospects that mostly outperform but infrequently extremely underperform—commonly referred to as “picking up pennies in front of a steamroller” (e.g., buying yield enhancement products, [Henderson and Pearson, 2011](#); [Célérier and Vallée, 2017](#); [Célérier et al., 2021](#); [Vokata, 2021](#); [Chesney et al., 2021](#)). Our framework predicts such heterogeneity in choice as a function of how people are exposed to information about the associated uncertainty. Finally, our framework provides a microfoundation for how heterogeneity in mental models can emerge *predictably* as a function of the information exposure method ([Kendall and Oprea, 2024](#); [Handel and](#)

⁴⁶[Manski \(2004\)](#) argues that choice data can be rationalized by multiple sets of preferences and beliefs. Thus the recovery of preferences from choice data requires either strong assumptions (e.g., rational expectations) or belief data. Our findings show that the assumption of rational expectations is violated even in settings where people have access to an objective distribution, and that this violation varies *predictably* with the way attention and memory interact with how participants are presented with information about the decision environment.

[Schwartzstein, 2018](#)).

Our model provides a common framework to link the literatures on sequential learning from personal experiences literature (e.g., ([Malmendier and Wachter, 2021](#))) and risky choice with “simultaneous” information about uncertainty. This common framework generates a rich set of predictions on how choices change due to the interaction between the information setting and cognitive constraints. It also provides an explanation for the “description-experience” gap in risky choice as stemming from the different way in which agents are exposed to information. Finally, while the majority of the literature on risky choice has focused on simultaneous information about uncertainty—and developed models to fit the observed behavior in such settings (e.g., prospect theory)—exposure to sequential information is an ecologically common way of learning about uncertainty. Thus understanding the link between the two has value in terms of connecting common models to real-world environments.

Lastly, our results suggest ways in which the decision environment can be tailored to improve the accuracy of mental representations. In our experiments, both the no recall bias and no salience bias settings lead to more accurate mental representations than the baseline settings. Relatedly, [Kaufmann et al. \(2013\)](#) document that a sampling procedure can be used to debias beliefs about returns of a single risky asset. In a context with multiple risky assets, [Laudenbach et al. \(2022\)](#) find that sequentially sampling return pairs instead of receiving simultaneous information about the joint return distribution leads to more accurate beliefs about dependence and makes portfolio choice more consistent with normative models ([Markowitz, 1952](#)). Our framework on how memory and attention affect choices could help structure the search for informational settings that generate accurate mental representations in specific contexts (see the literature on nudging and boosting, e.g. [Hertwig and Grüne-Yanoff, 2017](#)).

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A Theoretical Framework

A.1 Model of Saliency.

Following [Bruce and Tsotsos \(2009\)](#), the saliency of a state is a function of a bottom-up, stimulus-driven process that is part of sampling behavior. In their Attention based on Information Maximization (AIM) model, saliency emerges from first principles of bounded rationality, namely efficient coding and information theory. The saliency of some component (in our case, the state) is a function of the local information relative to the surrounding information; a state is more salient if the content is unexpected given the surrounding context.

For risky choice, an important factor that determines the degree to which a state departs from a local context are the associated outcomes. We extend the saliency framework of [Bordalo et al. \(2012\)](#) who capture a state’s saliency through the payoff outcomes that are associated with it by incorporating elements of local context. As shown in [Bruce and Tsotsos \(2009\)](#), spacial and visual representations are also an important aspect of saliency. A state whose outcomes depart strongly from the *local* payoffs of other states is more likely to be salient than a state that departs less from the local context.⁴⁷ Thus, our model takes saliency to be a function of (i) recalled outcomes for the given state and (ii) the contrast of those recalled outcomes—either payoff-based or visual—relative to the local context.⁴⁸

Let $\Omega^* \in \mathcal{O}(\Omega)$ denote the order in which states are displayed, where $\mathcal{O}(\Omega)$ denotes the set of all orderings of the state space, and let $\sigma : \mathbb{R}^{N \times K} \times \mathcal{O}(\Omega) \rightarrow \mathbb{R}^K$ denote a saliency function that maps the recalled outcomes and display order to a saliency value for each state. Thus $\sigma(\hat{X}, \Omega^*)$ denotes the saliency vector for perceived outcome matrix $\hat{X}$ and state order Ω^* . We write $\sigma(\omega; \hat{X}, \Omega^*)$ to denote the saliency of state ω . Assume σ is a continuous bounded function that satisfies the standard properties of a saliency function, as outlined here:

1. Ordering. Given state $\omega \in \Omega$, let $\hat{x}^{\min}(\omega)$ denote the lowest recalled outcome across all assets in state ω and let $\hat{x}^{\max}(\omega)$ denote the highest. Then given $\omega, \omega' \in \Omega$, if the range of recalled outcomes in state ω is a subset of the range in state ω' , $[\hat{x}^{\min}(\omega), \hat{x}^{\max}(\omega)] \subset [\hat{x}^{\min}(\omega'), \hat{x}^{\max}(\omega')]$ then state ω' is more salient:

$$\sigma(\omega; \hat{X}, \Omega^*) < \sigma(\omega'; \hat{X}, \Omega^*).$$

⁴⁷Other forms of visual stimuli include contrasting colors or pattern-based gestalt ([Mather, 2006](#)). [Ba et al. \(2022\)](#) explore how these forms of visual saliency impact beliefs in a belief updating framework.

⁴⁸Notably, as shown in [Bose et al. \(2022\)](#), [Li and Camerer \(2022\)](#), [Bruce and Tsotsos \(2009\)](#), amongst others, saliency is not not a free parameter. The saliency of an object can be predicted using algorithms such as the Graph-Based Visual Saliency model ([Harel et al., 2006](#)) or by tracking attention directly, as we show in [Section 5](#). Notably, the Graph-Based Saliency Model technique can be used to validate a model’s assumptions on what drives saliency in a particular setting, as well as to separate which aspects of saliency are driven purely by payoffs versus other types of visual stimuli (e.g., colors, relative placement in a field).

2. Diminishing sensitivity. If $\hat{x}^i(\omega) > 0$ for $i = 1, \dots, N$, then for any $\varepsilon > 0$,

$$\sigma(\omega; \hat{X}_\varepsilon, \Omega^*) < \sigma(\omega; \hat{X}, \Omega^*),$$

where $\hat{X}_\varepsilon$ denotes the matrix $\hat{X}$ with ε added to each element in the column corresponding to state ω .

3. Reflection. For any two states $\omega, \omega' \in \Omega$ such that $\hat{x}^i(\omega) > 0$ and $\hat{x}^i(\omega') > 0$ for $i = 1, \dots, N$,

$$\sigma(\omega; \hat{X}, \Omega^*) < \sigma(\omega'; \hat{X}, \Omega^*) \Leftrightarrow \sigma(\omega; -\hat{X}, \Omega^*) < \sigma(\omega'; -\hat{X}, \Omega^*).$$

4. Local Contrast. Consider two orderings of states, Ω_1^* and Ω_2^* . Suppose the recalled outcomes are the same across orderings, denoted by $\hat{X}$. Let $\eta(\omega, \Omega_k^*)$ denote the set of adjacent states to state ω given ordering Ω_k^* for $k = 1, 2$, and let $\bar{x}(\omega) = \frac{1}{N} \sum_{i=1}^N \hat{x}^i(\omega)$ denote the average recalled outcome in state ω . State ω is more salient in ordering Ω_1^* than Ω_2^* if its average outcome has a higher contrast relative to the average outcome of nearby states,

$$\frac{1}{|\eta(\omega, \Omega_1^*)|} \sum_{j \in \eta(\omega, \Omega_1^*)} |\bar{x}(\omega) - \bar{x}(\omega_j)| > \frac{1}{|\eta(\omega, \Omega_2^*)|} \sum_{j \in \eta(\omega, \Omega_2^*)} |\bar{x}(\omega) - \bar{x}(\omega_j)|$$

where $|\eta(\omega, \Omega_k^*)|$ denotes the number of adjacent states in ordering $k = 1, 2$.

We derive the salience ranking of states defined in [Section 2.1](#) from salience function $\sigma(\hat{X}, \Omega^*)$, where $k = (k(\omega_1), k(\omega_2), \dots, k(\omega_K))$ denotes the ranking of states by relative salience according to $\sigma(\hat{X}, \Omega^*)$.

A.2 Proofs of Predictions

Proof of Prediction 1. Assume $\delta < 1$. We start by showing that the salience of state ω_K is greater than the salience of any other state ω . Let X denote the matrix of outcomes for assets A^F and A^U and let $\Omega^* = (\omega_1, \dots, \omega_K)$ denote the ordering over states from lowest to highest payoff for A^U . Outcomes are correctly recalled in the simultaneous case, so $\hat{X} = X$. Therefore, for all $\omega \neq \omega_K$, $[\hat{x}^U(\omega), \hat{x}^F(\omega)] \subset [\hat{x}^F(\omega_K), \hat{x}^U(\omega_K)] = [x_1, x_K]$. By the ordering property of σ , this implies $\sigma(\omega_K; \hat{X}, \Omega^*) > \sigma(\omega; \hat{X}, \Omega^*)$. Therefore, state ω_K is the most salient: $k(\omega_K) = 1$ and $k(\omega) > 1$ for $\omega \neq \omega_K$.

Given that state ω_K is most salient, $\delta^{k(\omega_K)} > \delta^{k(\omega)}$ and it is outweighed:

$$\hat{\pi}(\omega_K) = \frac{\delta^{k(\omega_K)}}{\sum_{\omega' \in \Omega} \delta^{k(\omega')}} > \frac{\delta^{k(\omega)}}{\sum_{\omega' \in \Omega} \delta^{k(\omega')}} = \hat{\pi}(\omega), \quad (4)$$

where $\pi(\omega)$ cancels from the numerator and denominator since it is uniform. Since

the agent has a correct model of the outcome mapping,

$$\hat{E}[A^i] = \sum_{\omega \in \Omega} \hat{\pi}(\omega) * x^i(\omega) \quad (5)$$

$$V(A^i) = \sum_{\omega \in \Omega} \hat{\pi}(\omega) * v(x^i(\omega)) \quad (6)$$

for $i = U, F$. Given that both assets have the same set of outcomes across all states, $\sum_{\omega \in \Omega} v(x^U(\omega)) = \sum_{\omega \in \Omega} v(x^F(\omega))$. Let ω' denote the second most salient state. Then by the same reasoning as in (4), $\hat{\pi}(\omega') > \hat{\pi}(\omega)$ for all $\omega \in \Omega \setminus \{\omega_K, \omega'\}$. Then

$$\begin{aligned} V(A^U) - V(A^F) &= \sum_{\omega \in \Omega} \hat{\pi}(\omega) * (v(x^U(\omega)) - v(x^F(\omega))) \\ &= \hat{\pi}(\omega_K)(v(x^U(\omega_K)) - v(x^F(\omega_K))) + \sum_{\omega \neq \omega_K} \hat{\pi}(\omega)(v(x^U(\omega)) - v(x^F(\omega))) \\ &\geq \hat{\pi}(\omega_K)(v(x^U(\omega_K)) - v(x^F(\omega_K))) + \hat{\pi}(\omega') \sum_{\omega \neq \omega_K} (v(x^U(\omega)) - v(x^F(\omega))) \\ &= \hat{\pi}(\omega_K)(v(x^U(\omega_K)) - v(x^F(\omega_K))) - \hat{\pi}(\omega')(v(x^U(\omega_K)) - v(x^F(\omega_K))) \\ &= (\hat{\pi}(\omega_K) - \hat{\pi}(\omega'))(v(x^U(\omega_K)) - v(x^F(\omega_K))) > 0 \end{aligned}$$

where the third line follows from $v(x^U(\omega)) - v(x^F(\omega)) < 0$ for $\omega \neq \omega_K$, the fourth line follows from $\sum_{\omega \in \Omega} v(x^U(\omega)) = \sum_{\omega \in \Omega} v(x^F(\omega))$ which implies $\sum_{\omega \neq \omega_K} (v(x^U(\omega)) - v(x^F(\omega))) = -(v(x^U(\omega_K)) - v(x^F(\omega_K)))$, and the last line follows from $\hat{\pi}(\omega_K) > \hat{\pi}(\omega')$ and $v(x^U(\omega_K)) > v(x^F(\omega_K))$. Therefore, $V(A^U) > V(A^F)$. The analysis for $\hat{E}(A^U) - \hat{E}(A^F)$ follows immediately from taking $v(x) = x$. $\square$

Proof of Prediction 2. Consider the sequential environment. Symmetry of the similarity function implies $s^{UF} = s^{FU}$. Therefore, given that there are only two assets, the similarity vectors are equal, $s^U = s^F$. Thus we can write the recall probability for each asset as a function of s^{UF} : $r(2K, s^{UF}) = \min\{1, \alpha/(2K(1 + s^{UF}))\}$. There exists a number of states K_1 such that for $K > K_1$, $\alpha/(2K(1 + s^{UF})) < 1$. For the remainder of the proof, assume $K > K_1$. To simplify notation, denote $r(2K, s^{UF})$ as simply $r(K)$. Therefore, $r(K) = \alpha/(2K(1 + s^{UF})) \in (0, 1)$.

Prior to any memory distortions, $v(x^U(\omega)) - v(x^F(\omega)) < 0$ for $\omega \neq \omega_K$ and $v(x^U(\omega_K)) - v(x^F(\omega_K)) > 0$. Fix a state ω . If neither or only one outcome is misremembered in state ω , which happens with probability $(2 - r(K))r(K)$, then $v(\hat{x}^U(\omega)) - v(\hat{x}^F(\omega))$ has the same sign as and is (weakly) smaller in magnitude than $v(x^U(\omega)) - v(x^F(\omega))$. If both outcomes are misremembered, which happens with probability $(1 - r(K))^2$, then $\hat{x}^U(\omega) = \hat{x}^F(\omega) = (x^U(\omega) + x^F(\omega))/2$ and $v(\hat{x}^U(\omega)) - v(\hat{x}^F(\omega)) = 0$.

Turning to the recall process across all states, there are three cases:

1. The agent forgets both outcomes in state ω_K but remembers at least one out-

come in another state $\omega \neq \omega_K$. Then the recalled outcomes for asset A^F dominate the recalled outcomes for asset A^U : $\hat{x}^F(\omega) \geq \hat{x}^U(\omega)$ for all $\omega \in \Omega$ with the inequality strict for at least one state. Therefore, $V(A^F) > V(A^U)$ regardless of $\hat{\pi}$. This occurs with probability

$$p_F = (1-r(K))^2 - (1-r(K))^{2K} = \left(1 - \frac{\alpha}{2K(1+s^{UF})}\right)^2 - \left(1 - \frac{\alpha}{2K(1+s^{UF})}\right)^{2K}$$

2. The agent forgets both outcomes in all states. Then $\hat{x}^F(\omega) = \hat{x}^U(\omega)$ for all $\omega \in \Omega$ and $V(A^F) = V(A^U)$ regardless of $\hat{\pi}$. This occurs with probability

$$p_N = (1-r(K))^{2K} = \left(1 - \frac{\alpha}{2K(1+s^{UF})}\right)^{2K}.$$

3. The agent remembers either or both outcomes in state ω_K . Then whether $V(A^F) > V(A^U)$ or $V(A^F) < V(A^U)$ depends on the other recalled outcomes and $\hat{\pi}$. This occurs with probability

$$p_A = 2r(K) - r(K)^2 = \frac{2\alpha}{2K(1+s^{UF})} - \left(\frac{\alpha}{2K(1+s^{UF})}\right)^2$$

As $K \rightarrow \infty$,

$$p_F \rightarrow 1 - \exp(-\alpha/(1+s^{UF})) \quad (7)$$

$$p_N \rightarrow \exp(-\alpha/(1+s^{UF})) \quad (8)$$

$$p_A \rightarrow 0. \quad (9)$$

When $\alpha/(1+s^{UF}) > \ln 2$, $-\alpha/(1+s^{UF}) < \ln(0.5)$ and therefore $\exp(-\alpha/(1+s^{UF})) < 0.5$. Therefore, when $\alpha/(1+s^{UF}) > \ln 2$, there exists a number of states $\bar{K}$ such that for $K > \max\{K_1, \bar{K}\}$, $p_N < 0.5$, $p_F > 0.5$, and an expected utility maximizer is more likely to choose A^F than A^U . The same argument with $v(x) = x$ establishes the statement for $\hat{E}(A^U) - \hat{E}(A^F)$. $\square$

Proof of Prediction 3. Consider the sequential environment. Let $s^{UD} = S(A^U, A^D)$, $s^{UF} = S(A^U, A^F)$, and $s^{FD} = S(A^F, A^D)$. There exists a number of states K_1 such that for $K > K_1$, $\alpha/(2K(1+s^{UF})) < 1$ and therefore $r(2K, s^{UF}) = \min\{1, \alpha/(2K(1+s^{UF}))\} = \alpha/(2K(1+s^{UF})) < 1$. Suppose $K > K_1$ for the remainder of the proof. The probability the agent correctly recalls an outcome for A^F when A^D is present is weakly lower than the probability the agent correctly recalls an outcome for A^F in *Baseline Sequential*:

$$r(2K, (s^{UF}, s^{FD})) \leq r(2K, s^{UF}) \quad (10)$$

since $s^{UF} + s^{FD} \geq s^{UF}$ and $r(2K, s^i)$ is decreasing in $\sum_{j \neq i} s^{ij}$.⁴⁹ Similarly for recalling an outcome for A^U ,

$$r(2K, (s^{UF}, s^{UD})) \leq r(2K, s^{UF}) \quad (11)$$

since $s^{UF} + s^{UD} \geq s^{UF}$. Therefore, $r(2K, (s^{UF}, s^{FD})) = \alpha/(2K(1 + s^{UF} + s^{FD})) < 1$ and $r(2K, (s^{UF}, s^{UD})) = \alpha/(2K(1 + s^{UF} + s^{UD})) < 1$.

First consider the *Interference Asset U* environment (hereafter, *Interference*), with $s^{UD} > s^{UF} > s^{FD}$. When A^D is present, the agent correctly recalls an outcome for A^U with lower probability than for A^F :

$$r(2K, (s^{UF}, s^{UD})) < r(2K, (s^{UF}, s^{FD}))$$

since $s^{UF} + s^{UD} > s^{UF} + s^{FD}$ and $r(2K, s^i)$ is strictly decreasing in $\sum_{j \neq i} s^{ij}$. To simplify notation, let $r_B(K) = \alpha/(2K(1 + s^{UF}))$, $r_I^U(K) = \alpha/(2K(1 + s^{UF} + s^{UD}))$, and $r_I^F(K) = \alpha/(2K(1 + s^{UF} + s^{FD}))$ denote the probabilities of correct recall for either asset in *Baseline Sequential*, A^U in *Interference*, and A^F in *Interference*, respectively. By the above reasoning, $r_I^U(K) < r_I^F(K) \leq r_B(K) < 1$.

Since $s^{UD} > s^{UF}$, when the agent misremembers an outcome for A^U , in *Interference* she replaces it with an average of the outcomes for A^U and A^D and in *Baseline Sequential* she replaces it with an average of the outcomes for A^U and A^F . Therefore, given $x^D(\omega) > x^F(\omega)$, when she misremembers, her recalled outcome for A^U is higher in *Interference*:

$$\hat{x}_I^U(\omega) = \frac{x^U(\omega) + x^D(\omega)}{2} > \hat{x}_B^U(\omega) = \frac{x^U(\omega) + x^F(\omega)}{2}$$

where $\hat{x}_I^U$ denotes the recalled outcome distribution in *Interference* and $\hat{x}_B^U$ denotes the recalled outcome distribution in *Baseline Sequential*. Moreover, given $x^D(\omega) > x^U(\omega)$, in *Interference*, her recalled outcomes are higher when she misremembers than when she correctly recalls the outcome:

$$\frac{x^U(\omega) + x^D(\omega)}{2} > x^U(\omega).$$

Thus, the higher probability of misremembering in *Interference* increases her assessment of A^U .

Given $s^{UF} > s^{FD}$, when the agent misremembers an outcome for A^F , she replaces it with an average of the outcomes for assets A^F and A^U regardless of whether A^D

⁴⁹Note that by writing $r(2K, (s^{UF}, s^{FD}))$ instead of $r(3K, (s^{UF}, s^{FD}))$ we are implicitly assuming that only assets A^U and A^F affect the memory load (i.e., NK), as these are the only assets that the agent needs to recall at the time of choice. This assumption is innocuous: the result also holds if we assume that A^D impacts the memory load, since increasing the memory load exacerbates the effect of inference.

is present:

$$\hat{x}_I^F(\omega) = \hat{x}_B^F(\omega) = \frac{x^U(\omega) + x^F(\omega)}{2}.$$

Therefore, her recalled outcomes for A^F are unaffected by the presence of A^D .

Turning to the recall process across all states, there are three cases:

1. The agent forgets both outcomes in state ω_K but remembers at least one outcome in another state $\omega \neq \omega_K$. In *Interference*, it is possible that $V(A^F) > V(A^U)$ or $V(A^F) < V(A^U)$ depending on the other recalled outcomes and $\hat{\pi}$. This occurs with probability

$$p_{F,I} = (1 - r_I^U(K))(1 - r_I^F(K)) - (1 - r_I^U(K))^K(1 - r_I^F(K))^K.$$

As shown in the proof of [Prediction 2](#), in *Baseline Sequential*, the recalled outcomes for A^F dominate the recalled outcomes for A^U and $V(A^F) > V(A^U)$ regardless of $\hat{\pi}$. This occurs with probability

$$p_{F,B} = (1 - r_B(K))^2 - (1 - r_B(K))^{2K}.$$

2. The agent forgets both outcomes in all states. In *Interference*, $\hat{x}_I^U(\omega) > \hat{x}_I^F(\omega)$ for all $\omega \in \Omega$ and $V(A^U) > V(A^F)$ regardless of $\hat{\pi}$. This occurs with probability

$$p_{N,I} = (1 - r_I^U(K))^K(1 - r_I^F(K))^K.$$

As shown in the proof of [Prediction 2](#), in *Baseline Sequential*, $V(A^F) = V(A^U)$ regardless of $\hat{\pi}$ and this occurs with probability

$$p_{N,B} = (1 - r_B(K))^{2K}.$$

3. The agent remembers either or both outcomes in state ω_K . In both *Interference* and *Baseline Sequential*, it is possible that $V(A^F) > V(A^U)$ or $V(A^F) < V(A^U)$ depending on the other recalled outcomes and $\hat{\pi}$. In *Interference*, this occurs with probability

$$p_{A,I} = r_I^F(K) + r_I^U(K) - r_I^F(K)r_I^U(K)$$

and in *Baseline Sequential*, with probability

$$p_{A,B} = 2r_B(K) - r_B(K)^2.$$

From the proof [Prediction 2](#), we know that as $K \rightarrow \infty$,

$$p_{F,B} \rightarrow p_{F,B}^\infty := 1 - \exp(-\alpha/(1 + s^{UF})), \quad (12)$$

$$p_{N,B} \rightarrow p_{N,B}^\infty := \exp(-\alpha/(1 + s^{UF})), \quad (13)$$

$$p_{A,B} \rightarrow 0. \quad (14)$$

Similarly, in *Interference*, as $K \rightarrow \infty$,

$$\begin{aligned} p_{F,I} &= \left(1 - \frac{\alpha}{2K(1 + s^{UF} + s^{UD})}\right) \left(1 - \frac{\alpha}{2K(1 + s^{UF} + s^{FD})}\right) \\ &\quad - \left(1 - \frac{\alpha}{2K(1 + s^{UF} + s^{UD})}\right)^K \left(1 - \frac{\alpha}{2K(1 + s^{UF} + s^{FD})}\right)^K \\ &\rightarrow p_{F,I}^\infty := 1 - \exp\left(-\frac{\alpha}{2} \left(\frac{1}{1 + s^{UF} + s^{UD}} + \frac{1}{1 + s^{UF} + s^{FD}}\right)\right), \end{aligned} \quad (15)$$

$$\begin{aligned} p_{N,I} &= \left(1 - \frac{\alpha}{2K(1 + s^{UF} + s^{UD})}\right)^K \left(1 - \frac{\alpha}{2K(1 + s^{UF} + s^{FD})}\right)^K \\ &\rightarrow p_{N,I}^\infty := \exp\left(-\frac{\alpha}{2} \left(\frac{1}{1 + s^{UF} + s^{UD}} + \frac{1}{1 + s^{UF} + s^{FD}}\right)\right), \end{aligned} \quad (16)$$

$$\begin{aligned} p_{A,I} &= \frac{\alpha}{2K(1 + s^{UF} + s^{UD})} + \frac{\alpha}{2K(1 + s^{UF} + s^{FD})} \\ &\quad - \frac{\alpha}{2K(1 + s^{UF} + s^{UD})} \frac{\alpha}{2K(1 + s^{UF} + s^{FD})} \rightarrow 0 \end{aligned} \quad (17)$$

Note that $\exp(-\frac{\alpha}{2}\phi)$ is decreasing in ϕ for $\phi > 0$. Therefore, $p_{N,B}^\infty < p_{N,I}^\infty$ and $p_{F,B}^\infty > p_{F,I}^\infty$.

In *Interference*, the probability that $V(A^U) > V(A^F)$ and A^U is chosen is at least $p_{N,I}$. In *Baseline Sequential*, the probability that $V(A^U) > V(A^F)$ or $V(A^U) = V(A^F)$ is no greater than $p_{N,B} + p_{A,B}$. Thus, regardless of the tie-breaking rule when the agent is indifferent, the probability A^U is chosen is no greater than $p_{N,B} + p_{A,B}$ in *Baseline Sequential*. By the above limits, we can choose K sufficiently large so that $p_{N,I} > p_{N,B} + p_{A,B}$. Thus, the probability that the perceived expected utility of A^U exceeds that of A^F is strictly higher in *Interference* and therefore, an expected utility maximizer chooses A^U with strictly higher probability in *Interference*. The same argument with $v(x) = x$ establishes the statement for $\hat{E}(A^U) - \hat{E}(A^F)$. The proof for the case of *Interference Asset F* is analogous. $\square$

B Discussion of Additional Results

We discuss additional figures and tables from [Appendices C](#) and [D](#) here. [Appendix B.1](#) contains a few other results for our Baseline Experiment. [Appendices B.2](#) and [B.3](#) discuss the results from two additional experiments that explore variations in information exposure methods and the robustness of the effect of exposure methods on representations.

B.1 Baseline Experiment – Other Results

We report main results for our Baseline Experiment in [Tables 1](#) and [2](#) under (SQ1) and (SI1), as well as Figures [Fig. 4](#) and [Fig. 5](#). This appendix discusses some additional results, in particular the effect of information exposure methods on the strength of preferences for the chosen asset, splits by the incentivization scheme (choice vs. representations),⁵⁰ results for the long 51-state round of the experiment and persistence in choices across rounds, as well as choices conditional on representations. The summary statistics for the Baseline Experiment are in Panel A of [Table D.1](#).

B.1.1 Choices

We report other results on choices for our Baseline Experiment in [Table D.4](#). The first subpanel “All” combines all participants and rounds in each of the two presentation formats, including the 11-state distribution we focus on in the main paper and the 51-state distribution participants saw in the other round. Again, participants significantly favor the frequently-outperforming asset after the sequential information environment (83.54%, $p < .01$), but the unlikely-outperforming asset after the simultaneous information environment (57.22%, $p < .01$).

In this table, we also report the average self-assessed strength of preference for the participant’s choice. It is high (always above 4.8 on a scale from 1 to 7) in both exposure methods but systematically higher after sequential information. Participants in the simultaneous information environment have a systematically weaker preference for their choice (averages around 5 out of 7, compared to 6 out of 7 for the sequential information environment). The number of participants indicating indifference was quite low, corresponding to 5% of choices and did not differ between conditions. Removing indifferent choices does not meaningfully change the results neither qualitatively nor statistically.

The higher strength of preferences in the sequential information environment is inconsistent with an ambiguity-based explanation for choices after sequential information. In particular, one could argue that sequential information makes it more difficult for participants to understand the true distribution, even if they are told that the sample is representative, leading to more estimation uncertainty, less con-

⁵⁰See [Figs. C.5](#) to [C.8](#) for screenshots of the four variations of choice/representations elicitation screens in the sequential vs. simultaneous information environment.

fidence, or more ambiguity. In contrast, the simultaneous information environment could be argued to reduce ambiguity to zero, since participants directly get all relevant information about the distribution. However, participants state a significantly lower strength of preference with simultaneous information. This is in line with sequential information being a more natural information exposure method for most participants. Maybe this is not surprising given that most people (outside subpopulations in academia) are not used to absorb descriptions of probability distributions, whereas everybody is used to learning about uncertainty from sequential information (experience).

When splitting out participants who first made their choice and were incentivized for their choice (not their representations), results are virtually the same in the sequential information environment. The result for the simultaneous information environment is a bit weaker (54.64%, $p > .10$), but not statistically significantly different and it points in the same direction. Accordingly, the simultaneous information result is stronger (59.89%, $p < .01$) in the group of participants who were first asked for their representations and incentivized for accurate representations (and not their choice).

When splitting out the long 51-state round, we find qualitatively the same results as for the 11-state round reported in the main paper. Learning in the sequential information environment is followed by significantly more choices for the frequently-outperforming asset whereas people select the unlikely-outperforming asset significantly more often in the simultaneous information environment.

There is strong persistence between rounds in the simultaneous information environment. The majority of participants who select the unlikely-outperforming asset after the simultaneous information environment in round 1 (230 of 381) tend to also do so in round 2 (68.70%, $p < .01$). The minority of participants who select the frequently-outperforming asset in the simultaneous information environment in round 1 (151 of 381) tend to also do so in round 2 (68.21%, $p < .01$). We do not find this predictable heterogeneity for participants in the sequential information environment. Even the small minority of participants who select the unlikely-outperforming asset in round 1 (71 of 404) tend to select the other asset in round 2 (67.61%, $p < .01$).

B.1.2 Representations

We report other results on representations for our Baseline Experiment in [Table D.5](#). Overall, results are consistent with the 11-state version of *Baseline* presented in the main paper. After learning in the sequential information environment, participants are more optimistic about the frequently-outperforming asset. After the simultaneous information environment, they are more optimistic about the unlikely-outperforming asset. That is, belief patterns are consistent with choices from [Table D.4](#).

Incentivizing participants for representations does not lead to less biased repre-

sentations. Relative optimism for the unlikely-outperforming asset in the simultaneous information environment does not decrease at all, but rather increases (driven by more accurate representations about the other, frequently-outperforming asset though).

Ex-ante, one could have expected the overestimation of the average to shrink significantly in the 51-state round relative to the 11-state round. After all, frequent states are characterized by the frequently outperforming asset having a \$0.04 higher outcome instead of the \$0.20 higher outcome from the 11-state round. Nevertheless, participants expect a \$0.13 outperformance by the frequently better asset on average (versus \$0.16 in the 11-state round), which is more than the *maximum* outperformance of the frequently-outperforming asset across all 51 states.

B.1.3 Representations and Choices

Last we analyze choices conditional on representations. This is not quite as clean as other tests, because we are splitting by participants' arguably noisy estimates for their representations, which could introduce biases. Nevertheless, [Table D.6](#) shows a strong correspondence between stated representations and choices. Participants who are more optimistic for an asset are much more likely to select that alternative. Of course, we ask each participant for both representations *and* choices, so that participants might be adjusting representations to be consistent with choices or vice versa. Experiment A2 rules out this channel by asking half of participants only for representations, and the other half only for a choice.

B.2 Experiment A1 – Robustness with other Presentation Formats

In Experiment A1, we test variations of the simultaneous information environment (tabular simultaneous information without a graph, purely verbal simultaneous information without a graph) and a variation where participants view not just outcome levels for both assets but also the difference (which may be argued to make extreme outcome pairs more salient).

Experiment A1 is programmed in oTree ([Chen et al., 2016](#)). Like the experiments in the main paper, it is run online, using Prolific ([Gupta et al., 2021](#)) to recruit participants from the US/UK. However, we do not select the investor filter on Prolific, and the experiment does not include bot and attention check at the beginning, or a comprehension check before round 1. Summary statistics for participants in Experiments A1 are reported in Panel A of [Table D.7](#). This experiment was run in November 2022 and preregistered on AsPredicted.org, see [this link](#). Including a base fee, the average hourly salary for the experiment is more than \$20.

The design of Experiment A1 is very similar to the Baseline Experiment, but there are some differences. We now list the essential differences. Detailed instructions are reported in a supplementary document provided under [this link](#).

First, Experiment A1 always first asks participants to make their choice between the two risky alternatives. Participants are subsequently also asked to state their representations for the two alternatives, but representations are not incentivized. In short, we do not include the representations group from the Baseline Experiment.

Second, there are two simultaneous information environments In Experiment A1, which are both different from the Baseline Experiment’s simultaneous information environment. The main difference is the absence of a graph so that there is hardly any room for visual salience to matter. Some participants view a purely verbal simultaneous information format, similar to classical decision science research, see [Fig. C.9](#). Others view a tabular simultaneous information format, which conveniently lists all outcome pairs sorted by one of the two assets’ outcomes, see [Fig. C.10](#).

Third, for both the sequential and simultaneous information environment, some participants see not just levels of outcome variables, but also differences. See [Figs. C.11](#) and [C.12](#) for the verbal and tabular simultaneous information environments including differences, and [Fig. C.13](#) for the sequential information environment with differences. [Fig. C.14](#) shows the sequential information environment without differences, which is nearly identical to the Qualtrics version from our Baseline Experiment.

B.2.1 Choices

[Table D.8](#) shows our main result for Experiment A1. As in our Baseline Experiment’s results, we find that more than 2 out of 3 participants favor the frequently-outperforming asset after learning in the sequential information environment. While the effect is large, it is not quite as large as for the Baseline Experiment (72.95% instead of 83.54% select the frequently-outperforming asset), which may be due to the inclusion of language and bot checks, comprehension questions, as well as the investor filter in the Baseline Experiment. For participants in both simultaneous information environments, choices are statistically insignificantly different from 50% instead of being tilted towards the unlikely-outperforming alternative as in the Baseline Experiment. This may be due to the minor role for visual salience in Experiment A1’s non-graphical formats.

Overall, we also find a highly significant choice gap in Experiment A1. In line with our model and bounded memory as the driving force after sequential information, participants favor the frequently-outperforming asset in the sequential information environment. The lack of visually salient extreme states in both the verbal and the tabular simultaneous information environments is consistent with choices that do not significantly differ from 50% in favor of either asset.

When splitting out participants who view not just levels of outcomes, but also differences, we observe qualitatively the same results. Similarly, there is no meaningful difference between the 11-state distribution in this experiment and the 101-statedistribution. The very lengthy sequential sampling procedure for the 101 out-

come pairs in the long round might explain a slightly weaker choice pattern in that subsample compared to the 11-state distribution.

Like in our Baseline Experiment, we find choices in the simultaneous information environment to be persistently heterogeneous. While participants do not favor one or the other asset on average, it is possible to use round 1 to identify those who tend to select the frequently-outperforming or the unlikely-outperforming asset. This is not the case for the sequential information environment, where even the few participants who selected the unlikely-outperforming asset in round 1 are significantly more likely to select the frequently-outperforming asset in round 2.

B.2.2 Representations

As for the Baseline Experiment, we find that participants’ representations are consistent with their choices. In the sequential information environment, they are more optimistic about the frequently-outperforming alternative, see [Table D.9](#), both for the expected outcome (Panel A) and their risk assessment (Panel B).⁵¹ Like for choices, representations are not significantly different from the neutral response in the simultaneous information environment.

[Table D.10](#) shows a strong correspondence between stated representations and choices. Participants who are more optimistic for an alternative tend to select that alternative. Of course, representations are not incentivized and elicited after the choice, which could drive participants to adjust their representations so that they are consistent with their choice. Experiment A2 rules out that these design elements of Experiment A1 drive the evidence for a representations-based mechanism.

B.3 Experiment A2 – Isolating Biased Representations

In Experiment A2, we make sure that our results are not driven by spill-over effects of one task (e.g., the choice) to the incentivized main task (e.g., the representations elicitation). We either ask participants only to make an incentivized choice or only to state incentivized representations. Additionally, some participants are also presented with an information treatment (summarizing characteristics of the distribution of outcomes like equal marginal distributions) in an attempt to de-bias representations.

Experiment A2 is programmed in oTree ([Chen et al., 2016](#)). Like the experiments in the main paper, it is run online, using Prolific ([Gupta et al., 2021](#)) to recruit participants from the US/UK. In contrast to the Baseline Experiment, we do not select the investor filter on Prolific, and the experiment does not include bot and attention check at the beginning, or a comprehension check before round 1. Summary statistics for participants in Experiments A2 are reported in Panel B of [Table D.7](#). This experiment was run in March 2023 and preregistered on AsPredicted.org, see

⁵¹In contrast to our Baseline Experiment’s more involved probability estimates, we had participants score how risky each alternative is on a simple scale from 1 to 7.

[this link](#). Including a base fee, the average hourly salary for the experiment is more than \$20.

The design of Experiment A2 is very similar to the Baseline Experiment and Experiment A1. However, there are some differences. We now list the essential differences. Detailed instructions are reported in a supplementary document provided under [this link](#).

First, to reduce the number of variations compared to Experiment A1, we drop the difference formats from Experiment A1, which showed the difference between outcomes on top of the levels of outcomes for each situation. We also drop the tabular simultaneous information environment. We still use Experiment A1’s verbal simultaneous information environment, which is different from the Baseline Experiment (see [Appendix B.2](#)).

Second, in contrast to Experiment A1, half of participants are only asked to state their representations, without any mention of a choice between the two risky alternatives. Accurate representations are incentivized, see [Fig. C.15](#), like in our Baseline Experiment. Consistent with our Baseline, we also elicit representations about the probability distribution, on top of representations about the expected outcome, instead of just a risk score like in Experiment A1, see [Fig. C.16](#). The other half of participants are only asked to make a choice, without any mention of representations. They are incentivized for their choice, as in Experiment A1. This avoids any influence of this secondary task on the main task.

Third, half of participants in each exposure method (sequential and simultaneous information) and task (representations and choice) additionally see an information screen, which verbally describes key features of the joint distribution, such as equal marginal distributions and the existence of an infrequent but extreme outcome pair, see [Fig. C.17](#). One could argue that such a description should de-bias representations about marginal distributions, which could then be used to analyze the causal effect of representations on choices.

B.3.1 Choices

[Table D.11](#) confirms our main result from Experiment A1. The sequential information environment leads to a strong attraction towards the frequently-outperforming alternative. There is a highly significant choice gap, with choices based on simultaneous information not deviating significantly from 50%. The information screen does not seem to affect choices significantly, similar to the 11-state versus 51-state variations. We replicate our results from Experiment A1, although representations are never mentioned to participants in the choice group.

Consistent with our Baseline Experiment and Experiment A1, there is persistent heterogeneity in choices in the simultaneous information environment, though this persistence is less significant compared to the other experiments. Participants who

select the unlikely-outperforming asset in the simultaneous information environment in round 1 tend to also do so in round 2, and vice versa for participants who select the frequently-outperforming asset in round 1. Again, we do not find this predictable heterogeneity for participants in the sequential information environment.

In line with our Baseline Experiment, we again find that sampling the extreme outcome pair leads to a significant increase in viewing times. [Table D.12](#) shows the viewing time regressions for Experiment A2 and indicates that viewing times for the average draw increase from 1.62 seconds for a moderate pair to 2.76 seconds for an extreme pair, by 1.15 seconds ($p < .01$). So, extreme outcome pairs are attention-grabbing, but like in our Baseline Experiment we do not find that increased average attention towards the extreme pair changes average choices.

[Fig. C.18](#) splits participants into the many who pay overproportional versus the few who pay underproportional attention towards the extreme outcome pair. The main result does not show a meaningful difference. In [Table D.13](#), we report viewing time regressions, which do show statistically significant effects. Like in our Baseline Experiment, these are economically too small to meaningfully affect the main result, even for participants who pay many standard-deviations more attention towards the extreme outcome pair than the average participant.

B.3.2 Representations

As in our Baseline Experiment and Experiment A1, representations remain overoptimistic for the frequently better alternative in the sequential information environment (see [Table D.14](#)). This is despite participants only stating representations (not making a choice) and being incentivized for accurate representations. Like for choices, representations are not significantly different from the neutral response in the simultaneous information environment.

We find that the information intervention is ineffective in debiasing representations. This is consistent with choices in [Table D.11](#) not reacting to the information treatment. More generally, it is consistent with the small effect non-graphical (verbal or tabular) simultaneous information environments in Experiments A1 and A2 have on choices and representations.

C Additional Figures

Figure C.1: Flow of the experiments. This figure illustrates the flow of our Baseline Experiment. It begins with instructions, which include information on the upcoming tasks (choices and representation elicitation), information to come (presentation of risky alternatives), as well as an explanation of how participants are paid (performance-based incentivization either for their choice or stated representations). Subsequently participants go through two rounds, where they choose between two alternatives and state representations. Specifically, they are first shown information on the outcomes of two risky alternatives (or ‘assets’). The information exposure method is randomly allocated between participants, including both simultaneous information environments (stating outcomes for all possible states and the probabilities) and sequential information environments (where participants draw a representative sample from the underlying joint distribution). Then, participants make a choice between the assets and/or report their representations about the expected outcomes and likelihoods of outcomes. After finishing the second round, participants are asked a few concluding questions, before they are informed about their payment.

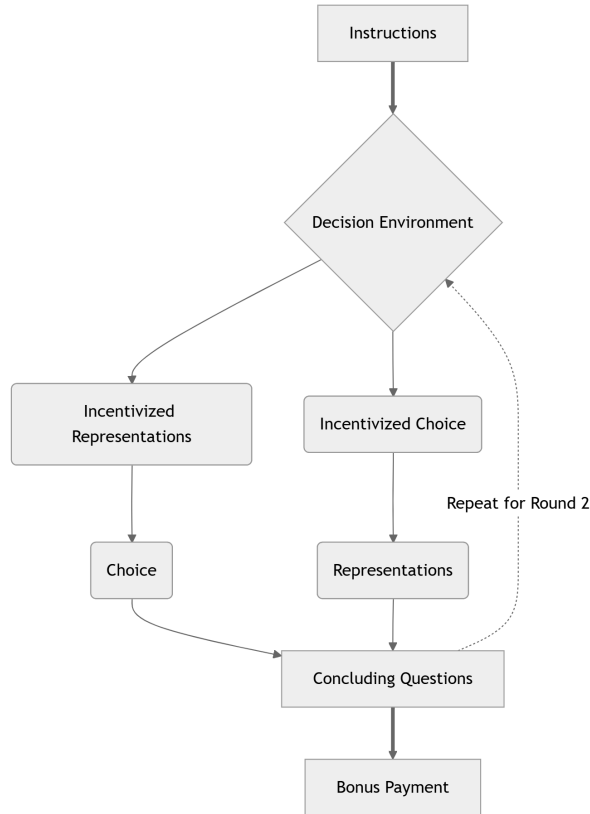


Figure C.2: Baseline Experiment: Simultaneous Information Environment for the 11-State Round. This figure shows two versions of the simultaneous information graph, which is used in our Baseline Experiment to present participants in the simultaneous information exposure method with the two risky assets when they make choices / state representations (see ‘Presentation of Risky Assets’ in Figure C.1). Neither choices nor representations differed significantly between the two versions, so results from both are combined in the analyses. The graphical and verbal description are analogous in the 51-state round. This graph is shown shown in the simultaneous information environment on the choice / representations screens.

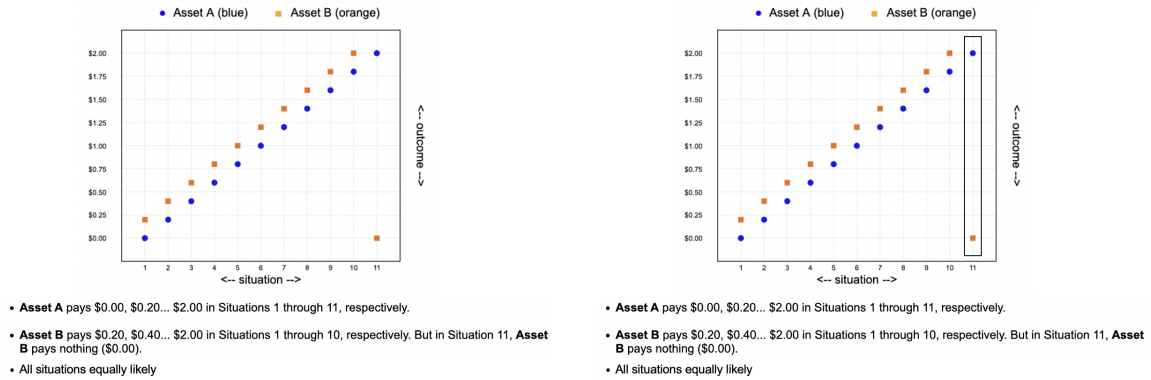


Figure C.3: Sequential Information with “No Recall Bias” – Change from Baseline. This figure illustrates the changes we made for the Sequential Information treatment in the “No Recall Bias” condition.

Panel A: Early Announcement Screen

You will be asked to write down and keep track of some of the information that you see in this experiment.
Please get a pen and paper or other method of writing down the information now.

Panel B: Detailed Explanation Before Sequential Information

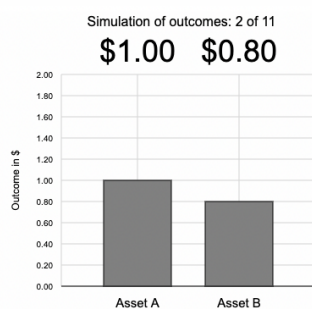
We will now begin the experiment. In this round, there are two Assets A and B and 11 possible situations, which are equally likely. On the screens that follow, you will see a simulation of how each Asset does in each of the 11 possible situations. This will give you an impression of how the assets perform. The outcomes are drawn from the joint distribution of outcomes across the two assets.

The bonus will be based on the accuracy of one of your guesses. The guess for your payment will be randomly selected.

The graph below depicts a typical simulation screen. In this situation, Asset A generated an outcome of 1.00 while Asset B generated an outcome of 0.80. You will see a graph like this across all 11 possible situations.

Please write down the outcomes for every single situation that you are shown, 11 total. For example, in the situation below, you can write down “Situation 2, Asset A returned \$1.00 and Asset B returned \$0.80.” You will be asked to recall this information at the end of the study and be paid a bonus of \$1 if you recall this information correctly.

You will then make decisions based on this information.



Panel C: Opportunity for Bonus After Choice & Representations

For each situation, please report below which asset returned more in that situation (or if both assets returned the same amount). If you answer each question correctly, you will receive a bonus of \$1.

	Asset A returned more than Asset B.	Asset B returned more than Asset A.	Both assets returned the same amount.
Situation 1	☐	☐	☐
Situation 2	☐	☐	☐
Situation 3	☐	☐	☐
Situation 4	☐	☐	☐
Situation 5	☐	☐	☐
Situation 6	☐	☐	☐
Situation 7	☐	☐	☐
Situation 8	☐	☐	☐
Situation 9	☐	☐	☐
Situation 10	☐	☐	☐
Situation 11	☐	☐	☐

Figure C.4: This figure reports results from the four-fold pattern experiment in Section 6.1. The panel on the left includes results from the risky choice design. The panel on the right includes results from deterministic mirror version of the same design.

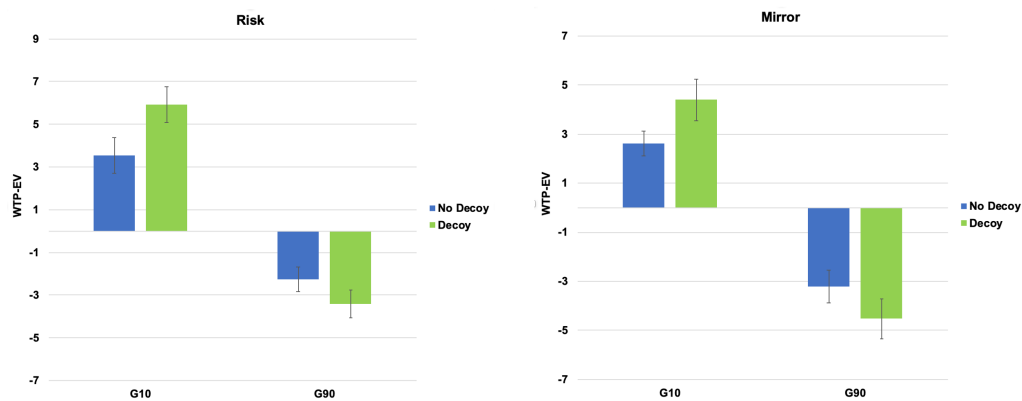


Figure C.5: Baseline Experiment: Elicitation of Choices and Representations in the Choice-Sequential Information Group. This figure shows the screens, which are used to ask participants who see the sequential information environment for their choice and representations (first asked for their choice, see Panel A, then for beliefs, see Panel B).

Panel A: Choice screen

Decisions

Based on the outcomes you have just seen, which Asset do you choose?

Asset B
☐

Asset A
☐

How strong is your preference for the asset you chose? Please select a category between 1 ("more or less indifferent") and 7 ("strong preference").

1
☐

2
☐

3
☐

4
☐

5
☐

6
☐

7
☐

Remember: If this round is selected for your bonus payment, the bonus will be based on a draw from your chosen asset's outcome distribution.

Panel B: Representations screen

Beliefs

Please guess the average outcome (in dollars) for each of the two assets:

Asset A
Average Outcome

Asset B

Please guess the chance (in %) that **Asset A** generates each of these three potential outcomes:

Outcome below 1.00	%
Outcome of exactly 1.00	%
Outcome above 1.00	%
Total	%

Please guess the chance (in %) that **Asset B** generates each of these three potential outcomes:

Outcome below 1.00	%
Outcome of exactly 1.00	%
Outcome above 1.00	%
Total	%

Please guess the chance (in %) that the outcome of Asset A is:

Better than asset B	%
Equal to asset B	%
Worse than asset B	%
Total	%

Note that you will not be able to proceed to the next page unless the probabilities add up to a total of 100%.

Figure C.6: Baseline Experiment: Elicitation of Representations and Choices in the Representations-Sequential Information Group. This figure shows the screens, which are used to ask participants who see the sequential information environment for their representations and choice (first asked for their representations, see Panel A, then for choice, see Panel B).

Panel A: Representations screen

Beliefs

Please guess the average outcome (in dollars) for each of the two assets:

	Asset A	Asset B
Average Outcome		

Please guess the chance (in %) that **Asset A** generates each of these three potential outcomes:

Outcome below 1.00	%
Outcome of exactly 1.00	%
Outcome above 1.00	%
Total	%

Please guess the chance (in %) that **Asset B** generates each of these three potential outcomes:

Outcome below 1.00	%
Outcome of exactly 1.00	%
Outcome above 1.00	%
Total	%

Please guess the chance (in %) that the outcome of Asset A is:

Better than asset B	%
Equal to asset B	%
Worse than asset B	%
Total	%

Note that you will not be able to proceed to the next page unless the probabilities add up to a total of 100%.

Remember: If this round is selected for your bonus payment, the bonus will be based on the accuracy of one of your guesses. The guess for your payment will be randomly selected.

[→](#)

Panel B: Choice screen

Decisions

Based on the outcomes you have just seen, which Asset do you choose?

Asset A	Asset B
☐	☐

How strong is your preference for the asset you chose? Please select a category between 1 ("more or less indifferent") and 7 ("strong preference").

1	2	3	4	5	6	7
☐	☐	☐	☐	☐	☐	☐

[→](#)

Figure C.7: Baseline Experiment: Elicitation of Choices and Representations in the Choice-Simultaneous Information group. This figure shows the screens, which are used to ask participants who see the simultaneous information environment for their choices and representations (first asked for their choice, see Panel A, then for representations, see Panel B).

Panel A: Choice screen

Decisions

Based on the outcomes you see above, which Asset do you choose?

Asset A

☐

Asset B

☒

How strong is your preference for the asset you chose? Please select a category between 1 ("more or less indifferent") and 7 ("strong preference").

1
☐

2
☐

3
☐

4
☐

5
☐

6
☒

7
☐

Remember: If this round is selected for your bonus payment, the bonus will be based on a draw from your chosen asset's outcome distribution.

→

Panel B: Representations screen

Beliefs

Please guess the average outcome (in dollars) for each of the two assets:

Asset B

Asset A

Please guess the chance (in %) that **Asset A** generates each of these three potential outcomes:

Outcome below 1.00	%
Outcome of exactly 1.00	%
Outcome above 1.00	%
Total	%

Please guess the chance (in %) that **Asset B** generates each of these three potential outcomes:

Outcome below 1.00	%
Outcome of exactly 1.00	%
Outcome above 1.00	%
Total	%

Please guess the chance (in %) that the outcome of Asset A is:

Better than asset B	%
Equal to asset B	%
Worse than asset B	%
Total	%

Note that you will not be able to proceed to the next page unless the probabilities add up to a total of 100%.

Remember: If this round is selected for your bonus payment, the bonus will be based on the accuracy of one of your guesses. The guess for your payment will be randomly selected.

→

Figure C.8: Baseline Experiment: Elicitation of Representations and Choices in the Representations-Simultaneous Information group. This figure shows the screens, which are used to ask participants who see the simultaneous information environment for their representations and choice i (first asked for their representations, see Panel A, then for choice, see Panel B).

Panel A: Representations screen

Beliefs

Please guess the average outcome (in dollars) for each of the two assets:

	Asset B	Asset A
Average Outcome		

Please guess the chance (in %) that **Asset A** generates each of these three potential outcomes:

Outcome below 1.00	%
Outcome of exactly 1.00	%
Outcome above 1.00	%
Total	%

Please guess the chance (in %) that **Asset B** generates each of these three potential outcomes:

Outcome below 1.00	%
Outcome of exactly 1.00	%
Outcome above 1.00	%
Total	%

Please guess the chance (in %) that the outcome of Asset A is:

Better than asset B	%
Equal to asset B	%
Worse than asset B	%
Total	%

Note that you will not be able to proceed to the next page unless the probabilities add up to a total of 100%.

Remember: If this round is selected for your bonus payment, the bonus will be based on the accuracy of one of your guesses. The guess for your payment will be randomly selected.

[→](#)

Panel B: Choice screen

Decisions

Based on the outcomes you see above, which Asset do you choose?

Asset B	Asset A
☒	☐

How strong is your preference for the asset you chose? Please select a category between 1 ("more or less indifferent") and 7 ("strong preference").

1	2	3	4	5	6	7
☐	☐	☐	☐	☐	☒	☐

[→](#)

Figure C.9: Experiment A1: Verbal Simultaneous Information Environment.

This figure illustrates the information exposure method in the verbal simultaneous information group of Experiment A1, which is used to present participants with the two assets in the ‘Information’ stage (see dotted box in Figure C.1). The order of bullet points for Alternative 2 is randomized, so that the extreme outcome pair is randomly at the top or bottom. The allocation of the frequently-outperforming asset to Alternative 1 or 2 is also randomized.

Description of Outcomes

- Alternative 1 pays each of the 11 outcomes £0.00, £0.20, £0.40, ..., £2.00 with equal likelihood.
- Alternative 2 pays:
 - £2.00 when Alternative 1 pays £0.00 (with total probability 1/11).
 - £0.00, £0.20, £0.40, ..., £1.80 when Alternative 1 pays £0.20, £0.40, £0.60, ..., £2.00 (with probability 1/11 each, in total 10/11).

Next

Figure C.10: Experiment A1: Tabular Simultaneous Information Environment.

This figure illustrates the information exposure method in the tabular simultaneous information group of Experiment A1, which is used to present participants with the two assets in the ‘Information’ stage (see dotted box in Figure C.1). Outcome pairs are sorted either by Alternative 1 or 2, so that the extreme outcome pair is randomly at the top or bottom. The allocation of the frequently-outperforming asset to Alternative 1 or 2 is also randomized.

Description of Outcomes

All states are equally likely

State	Outcome for Alternative 1	Outcome for Alternative 2
1	£0.00	£2.00
2	£0.20	£0.00
3	£0.40	£0.20
4	£0.60	£0.40
5	£0.80	£0.60
6	£1.00	£0.80
7	£1.20	£1.00
8	£1.40	£1.20
9	£1.60	£1.40
10	£1.80	£1.60
11	£2.00	£1.80

Next

Figure C.11: Experiment A1: Verbal Simultaneous Environment with Differences. This figure illustrates the information exposure method in the verbal description group with differences of Experiment A1, which is used to present participants with the two assets in the ‘Information’ stage (see dotted box in Figure C.1). The order of bullet points for Alternative 2 is randomized, so that the extreme outcome pair is randomly at the top or bottom. The allocation of the frequently-outperforming asset to Alternative 1 or 2 is also randomized.

Description of Outcomes

- Alternative 1 pays each of the 11 outcomes £0.00, £0.20, £0.40, ..., £2.00 with equal likelihood.
- Alternative 2 pays:
 - £2.00 more than Alternative 1 when Alternative 1 pays £0.00 (with total probability 1/11). That means it pays £2.00.
 - £0.20 less than Alternative 1 when Alternative 1 pays £0.20, £0.40, £0.60, ..., £2.00 (with probability 1/11 each, in total 10/11). That means it pays £0.00, £0.20, £0.40, ..., £1.80.

Next

Figure C.12: Experiment A1: Tabular Simultaneous Information Environment with Differences. This figure illustrates the information exposure method in the tabular simultaneous information with differences group of Experiment A1, which is used to present participants with the two assets in the ‘Information’ stage (see dotted box in Figure C.1). Outcome pairs are sorted either by Alternative 1 or 2, so that the extreme outcome pair is randomly at the top or bottom. The allocation of the frequently-outperforming asset to Alternative 1 or 2 is also randomized.

Description of Outcomes

All states are equally likely

State	Outcome for Alternative 1	Difference	Outcome for Alternative 2
1	£0.00	+£2.00	£2.00
2	£0.20	-£0.20	£0.00
3	£0.40	-£0.20	£0.20
4	£0.60	-£0.20	£0.40
5	£0.80	-£0.20	£0.60
6	£1.00	-£0.20	£0.80
7	£1.20	-£0.20	£1.00
8	£1.40	-£0.20	£1.20
9	£1.60	-£0.20	£1.40
10	£1.80	-£0.20	£1.60
11	£2.00	-£0.20	£1.80

[Next](#)

Figure C.13: Experiment A1: Sequential Information Environment with Differences. This figure illustrates the information exposure method in the sequential information group with differences of Experiment A1, which is used to present participants with the two assets in the ‘Information’ stage (see dotted box in Figure C.1). Subjects sample T outcome pairs for the two assets (T=11 or 101 in Experiment A1’s two rounds, counterbalanced), before moving on to the tasks of the experiment (choice and belief elicitation). Outcome pairs make up a representative distribution and are sampled (without replacement) in random order for each participant. The allocation of the frequently-outperforming asset to Alternative 1 or 2 is also randomized.

Panel A: Illustration sequential information screen, moderate outcome pair



Panel B: Illustration summary screen, extreme outcome pair

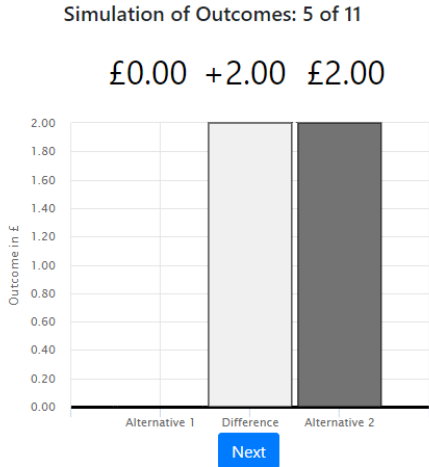


Figure C.14: Experiment A1: Sequential Information Environment. This figure illustrates the information exposure method in the sequential information group of Experiment A1, which is used to present participants with the two assets in the ‘Information’ stage (see dotted box in Figure C.1). Subjects sample T outcome pairs for the two assets (T=11 or 101 in Experiment A1’s two rounds, counterbalanced), before moving on to the tasks of the experiment (choice and belief elicitation). Outcome pairs make up a representative distribution and are sampled (without replacement) in random order for each participant. The allocation of the frequently-outperforming asset to Alternative 1 or 2 is also randomized.

Panel A: Illustration sequential information screen, moderate outcome pair



Panel B: Illustration summary screen, extreme outcome pair



Figure C.15: Experiment A2: Instructions Representations Group. This figure illustrates the instructions received by participants in the representations group of Experiment A2 before the start of the first round. The pop-up window in Panel B appears when participants click on ‘More info’.

Panel A: Main instructions

Bonus Payment

In the following, you are asked to guess outcomes for two different alternatives. You will receive more information about the alternatives soon.

Your bonus payment will be based on the accuracy of one of your guesses, which will be randomly drawn.

Your payment is calculated as follows: [Bonus Pay] + [Base Pay] = [Total Pay]

You can potentially earn an additional bonus of £2.00. At the end of the study, we will randomly select one of your guesses. The closer your guess is to the actual value, the higher is the likelihood of you receiving the bonus. In case you are interested, you can see the specific method that determines the bonus by clicking the "More Info" button.

Next

More Info

Panel B: Pop-up window

More Info

If your guess is within 5% of the statistically correct value, you will earn the additional £2.00. Therefore, in order to earn as much money as possible, you should try to give your best guess for each of the values being asked.

For example, if we ask for a probability and the statistically correct probability is 50%, then you will receive the bonus if you state anything between 47.5% and 52.5%. If we ask for an average outcome and the statistically correct average outcome is 1.00, you will receive the bonus if your guess is between 0.95 and 1.05.

Close

Figure C.16: Experiment A2: Representation Elicitation Screen. This figure shows the screen shown to participants in the representations group of Experiment A2 to ask for their beliefs about Alternatives 1 and 2.

Beliefs

Please guess the average outcome for each of the two alternatives:

Outcome	Alternative 1	Alternative 2
Average Outcome		

Please guess the probability (in %) that each of the two alternatives has one of these three outcomes:

Outcome	Alternative 1	Alternative 2
Outcome below 1.00		
Outcome of exactly 1.00		
Outcome above 1.00		
Total (in %):		

Remember: If this round is selected for your bonus payment, the bonus will be based on the accuracy of one of your guesses, which will be randomly drawn.

Note that you will not be able to proceed to the next page unless the probabilities add up to a total of 100%.

Figure C.17: Experiment A2: Information Screen. This figure shows the information screen shown to half of participants of Experiment A2 to facilitate learning about important characteristics of Alternatives 1 and 2. Both participants in the representations and the choice group are randomly selected to see this screen (or not) right before their main task (choice or representations).

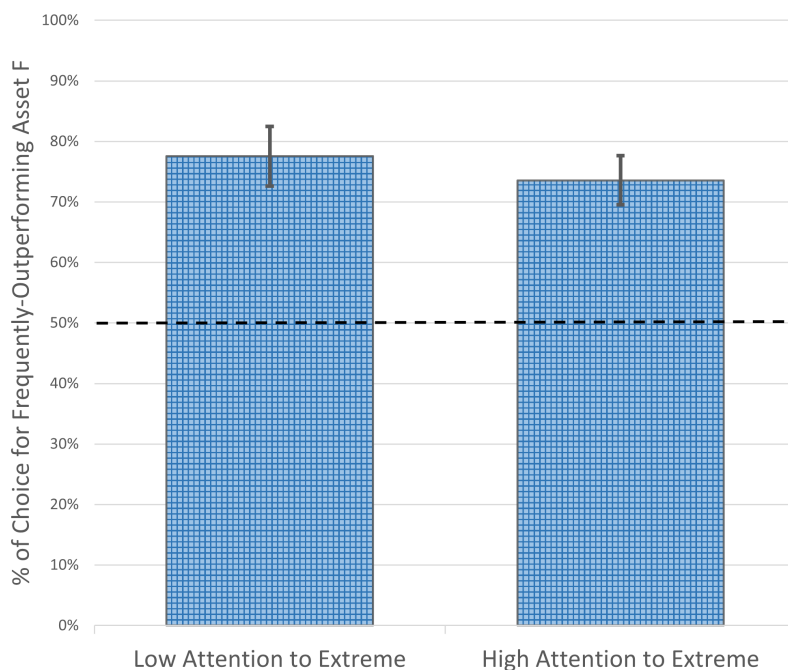
Information Screen

Based on the information you just received, you know the following about Alternatives 1 and 2:

- One alternative has a **slightly higher outcome** than the other alternative **most of the time** (with a high likelihood). However, the frequently better alternative has a **much lower outcome occasionally** (with a low likelihood).
- Considered **individually**, both alternatives have **precisely the same outcome distributions**. That means that their **average outcomes are the same**. So are the **likelihoods** that they realize an **outcome below any given threshold value**.

Next

Figure C.18: Experiment A2: Split conditional on viewing-time of extreme outcome pair. This figure shows the main outcome for participants in the choice sequential information group, split by whether their viewing time for the extreme outcome pair was higher or lower than their average viewing time for the moderate outcome pairs. The result for the 63% (37%) of participants with a viewing time for the extreme pair above (below) their average viewing time for the moderate pairs is shown on the right (left). In both cases, a significantly higher fraction of participants selects the frequently-outperforming alternative.



D Additional Tables

Table D.1: Summary statistics. This table reports summary statistics for our Baseline Experiment and the theory-testing experiments, which systematically manipulate memory and attention. ‘Age’ is the participant’s age in years and ‘Female’ is an indicator for the participant’s sex. Participants self-assess their willingness to take risks on a scale from 1 (take no risks) to 5 (take large risks to achieve a significant gain). Self-assessed statistical knowledge is based on a scale from 1 (poor) to 5 (excellent). The student indicator is from Prolific. Participants state, on a scale from 1 (never) to 5 (very often), whether they consider what could happen if they made another decision (ex-ante regret) or could have happened if they make another decision (ex-post regret).

Variable	Mean	Std.Dev.	p10	p50	p90
Panel A: Baseline Experiment: 785 participants (Investors, 73% from the UK, others from the US)					
Age	38.68	12.91	24	36	58
Female	0.50	0.50	0	1	1
Willingness to take risks (1-5)	3.23	0.98	2	3	4
Self-assessed stat. knowledge (1-5)	2.71	1.02	1	3	4
Student	0.12	0.33	0	0	1
Considers regret ex-ante (1-5)	4.00	0.87	3	4	5
Considers regret ex-post (1-5)	3.70	1.02	2	4	5
Time for full experiment (seconds)	885.08	448.47	486	755	1418
Panel B: Theory-Testing Experiments: 600 participants (Investors, 74% from the UK, others from the US)					
Age	37.00	11.90	24	34	55
Female	0.50	0.50	0	0	1
Willingness to take risks (1-5)	3.31	1.00	2	3	5
Self-assessed stat. knowledge (1-5)	2.78	1.05	1	3	4
Student	0.20	0.40	0	0	1
Considers regret ex-ante (1-5)	4.04	0.87	3	4	5
Considers regret ex-post (1-5)	3.89	0.89	3	4	5
Time for full experiment (seconds)	781.76	401.07	388	689	1312

Table D.2: Joint Distributions in Interference Treatments. This table shows a description of the joint distribution, which participants in *Interference Asset U* and *Interference Asset F* sampled. All 11 States are equally likely and participants view a representative sample in random order, just like in the baseline design. The two assets that are available in the choice exhibit the same marginal outcome distributions, but are first-order stochastically dominated by a new third asset, the decoy company, which is sampled to interfere with memory formation. In *Interference Asset U*, the decoy company is similar in color and industry to Asset U (Tech Company 2). Interference in memory formation should thus improve the perceived attractiveness of Asset U. In *Interference Asset F*, the decoy company is similar in color and industry to Asset F (Oil Company 2). Inference in memory formation should thus increase the attractiveness of Asset F. Note that the marginal distributions of the two available assets are slightly different from *Baseline Sequential*: outcomes are on an irregular grid to have more digits, which is necessary to avoid systematic differences relative to the decoy in the “roundness” of outcomes. The marginal distribution of Assets F and U are still the same and Asset F still outperforms Asset U by a moderate amount in 10 states and underperforms by a large margin in state 11.

	Asset D (decoy)	Asset U	Asset F
Interference: Asset U	Tech Company 1 light blue	Tech Company 2 dark blue	Oil Company 1 dark orange
Interference: Asset F	Oil Company 1 light orange	Tech Company 1 dark blue	Oil Company 2 dark orange
State 1	\$0.51	\$0.05	\$0.30
State 2	\$0.73	\$0.30	\$0.50
State 3	\$0.94	\$0.50	\$0.70
State 4	\$1.10	\$0.70	\$0.88
State 5	\$1.26	\$0.88	\$1.05
State 6	\$1.53	\$1.05	\$1.30
State 7	\$1.73	\$1.30	\$1.48
State 8	\$1.90	\$1.48	\$1.68
State 9	\$2.13	\$1.68	\$1.88
State 10	\$2.28	\$1.88	\$2.07
State 11	\$2.28	\$2.07	\$0.05

Table D.3: Baseline Experiment – Attention to Extremes. In this table, we report regressions of viewing times per outcome pair within the sequential information group of the Baseline Experiment on a dummy for the extreme outcome pair. Viewing time is measured in seconds (Specifications (1) to (3)) or log(seconds) (Specifications (4) to (6)). We split up all sampled outcome pairs into those sampled from short (11-pair) rounds versus long (51-pair) rounds in Specifications (2)-(3) and (5)-(6). t -statistics are based on participant-clustered standard errors. */**/** indicates statistical significance at the 10%/5%/1% level.

	All	Seconds Short	Long	All	Log(Seconds) Short	Long
$I_{\text{Extreme Pair}}$	1.780***	1.554***	2.009***	0.552***	0.446***	0.654***
	6.58	3.81	6.17	18.85	14.13	14.69
Constant	1.867***	2.855***	1.665***	0.154***	0.556***	-0.007***
	10.91	77.07	260.73	6.52	193.36	-8.08
R^2	12.36%	21.43%	13.38%	60.15%	71.09%	61.18%
N	25,048	4,444	20,604	25,032	4,443	20,589
Participant fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Round fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observation number fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Long round fixed effects	Yes	No	No	Yes	No	No
Clustered by participant	Yes	Yes	Yes	Yes	Yes	Yes

Table D.4: Baseline Experiment – Choices. In this table, we report average choices in our Baseline Experiment. We report the average fraction of participants selecting each of the alternatives. See Section B.1 for a more detailed description of the different conditions. t -statistics are based on participant-clustered standard errors whenever both rounds are included. */**/** indicates statistical significance at the 10%/5%/1% level.

Condition	Choice Asset U	Choice Asset F	Difference versus 50%	t -test	N	Strength of Preference (1-7)
All						
Sequential Information	16.46%	83.54%	33.54%***	23.28	808	5.84
Simultaneous Information	57.22%	42.78%	-7.22%***	-3.45	762	5.00
Choice Group						
Sequential Information	17.32%	82.68%	32.68%***	15.71	410	5.76
Simultaneous Information	54.64%	45.36%	-4.64%	-1.62	388	5.03
Representations Group						
Sequential Information	15.58%	84.42%	34.42%***	17.26	398	5.93
Simultaneous Information	59.89%	40.11%	-9.89%***	-3.24	374	4.97
Short round (11 outcome pairs)						
Sequential Information	15.10%	84.90%	34.90%***	19.57	404	5.72
Simultaneous Information	58.01%	41.99%	-8.01%***	-3.16	381	4.98
Long round (51 outcome pairs)						
Sequential Information	17.82%	82.18%	32.18%***	16.88	404	5.96
Simultaneous Information	56.43%	43.57%	-6.43%**	-2.53	381	5.02
Round 2 choice, when the round 1 choice was for the frequently-outperforming asset						
Sequential Information	11.71%	88.29%	38.29%***	21.70	333	5.87
Simultaneous Information	31.79%	68.21%	18.21%***	4.79	151	4.87
Round 2 choice, when the round 1 choice was for the unlikely-outperforming asset						
Sequential Information	32.39%	67.61%	17.61%***	3.15	71	5.69
Simultaneous Information	68.70%	31.30%	-18.70%***	-6.10	230	4.93

Table D.5: Baseline Experiment – Representations. In this table, we report average representations in our Baseline Experiment. Panel A reports the average outcome expected for each of the alternatives. Panel B reports the average probability of below/above-median outcomes for each of the alternatives. The t -statistics in Panel B are from a difference-in-difference estimate between the below- and above-median probability assessments. See Section B.1 for a more detailed description of the different conditions. t -statistics are based on participant-clustered standard errors whenever both rounds are included. */**/** indicates statistical significance at the 10%/5%/1% level.

Condition	Representations Asset U	Representations Asset F	Difference (-in-Diff.)	t -test	N
Panel A: Expected outcome (0-2)					
All					
Sequential Information	1.02	1.16	0.15***	9.10	808
Simultaneous Information	1.08	1.01	-0.07***	-3.76	762
Choice Group					
Sequential Information	1.01	1.19	0.18***	7.33	410
Simultaneous Information	1.08	1.05	-0.03	-1.32	388
Representations Group					
Sequential Information	1.02	1.13	0.11***	5.47	398
Simultaneous Information	1.08	0.98	-0.10***	-3.69	374
Short round (11 outcome pairs)					
Sequential Information	1.00	1.17	0.16***	7.34	404
Simultaneous Information	1.07	1.00	-0.07***	-2.83	381
Long round (51 outcome pairs)					
Sequential Information	1.03	1.16	0.13***	6.52	404
Simultaneous Information	1.09	1.02	-0.07***	-2.67	381
Panel B: Risk assessment (below/above-median probabilities)					
All					
Sequential Information – $P(x < 1)$	47.82%	37.48%	-20.44%***	-13.57	808
Sequential Information – $P(x > 1)$	40.06%	50.16%			
Simultaneous Information – $P(x < 1)$	39.53%	42.87%	6.07%***	4.04	762
Simultaneous Information – $P(x > 1)$	44.64%	41.91%			
Choice Group					
Sequential Information – $P(x < 1)$	48.43%	36.41%	-23.91%***	-10.24	410
Sequential Information – $P(x > 1)$	38.88%	50.78%			
Simultaneous Information – $P(x < 1)$	40.67%	42.50%	3.89%**	1.97	388
Simultaneous Information – $P(x > 1)$	45.24%	43.19%			
Representations Group					
Sequential Information – $P(x < 1)$	47.19%	38.59%	-16.85%***	-9.07	398
Sequential Information – $P(x > 1)$	41.28%	49.53%			
Simultaneous Information – $P(x < 1)$	38.35%	43.26%	8.34%***	3.67	374
Simultaneous Information – $P(x > 1)$	44.02%	40.59%			
Short round (11 outcome pairs)					
Sequential Information – $P(x < 1)$	48.40%	35.04%	-25.87%***	-12.58	404
Sequential Information – $P(x > 1)$	38.24%	50.76%			
Simultaneous Information – $P(x < 1)$	39.09%	42.63%	6.46%***	3.46	381
Simultaneous Information – $P(x > 1)$	44.27%	41.35%			
Long round (51 outcome pairs)					
Sequential Information – $P(x < 1)$	47.23%	39.92%	-15.00%***	-8.62	404
Sequential Information – $P(x > 1)$	41.88%	49.57%			
Simultaneous Information – $P(x < 1)$	39.96%	43.11%	5.68%***	3.12	381
Simultaneous Information – $P(x > 1)$	45.01%	42.48%			

Table D.6: Baseline Experiment – Representations and Choices. In this table, we report average choices in our Baseline Experiment, split based on their representations of the expected outcome and risk of alternatives. We report the average fraction of participants selecting each of the alternatives. See Section B.1 for a more detailed description of the different conditions. t -statistics are based on participant-clustered standard errors. */**/** indicates statistical significance at the 10%/5%/1% level.

Condition	Choice Asset U	Choice Asset F	Difference versus 50%	t -test	N
Representation in which frequently-outperforming asset has higher expected outcome					
Sequential Information	7.63%	92.37%	42.37%***	35.58	498
Simultaneous Information	42.04%	57.96%	7.96%**	2.01	157
Representation in which frequently-outperforming asset has lower expected outcome					
Sequential Information	29.83%	70.17%	20.17%***	5.91	181
Simultaneous Information	76.44%	23.56%	-26.44%***	-8.97	208
Representation in which frequently-outperforming asset has same expected outcome					
Sequential Information	31.78%	68.22%	18.22%***	4.43	129
Simultaneous Information	53.15%	46.85%	-3.15%	-1.26	397
Representation in which frequently-outperforming asset has lower risk					
Sequential Information	6.68%	93.32%	43.32%***	38.52	494
Simultaneous Information	46.15%	53.85%	3.85%	0.92	143
Representation in which frequently-outperforming asset has higher risk					
Sequential Information	43.61%	56.39%	6.39%	1.48	133
Simultaneous Information	75.20%	24.80%	-25.20%***	-9.28	254
Representation in which frequently-outperforming asset has same risk					
Sequential Information	23.20%	76.80%	26.80%***	8.52	181
Simultaneous Information	49.04%	50.96%	0.96%	0.37	365

Table D.7: Summary statistics. This table reports summary statistics for Experiments A1 and A2. ‘Age’ is the participant’s age in years and ‘Female’ is an indicator for the participant’s sex. Participants self-assess their willingness to take risks on a scale from 1 (take no risks) to 5 (take large risks to achieve a significant gain). Self-assessed statistical knowledge is based on a scale from 1 (poor) to 5 (excellent). The student indicator is from Prolific. Participants state, on a scale from 1 (never) to 5 (very often), whether they consider what could happen if they made another decision (ex-ante regret) or could have happened if they make another decision (ex-post regret).

Variable	Mean	Std.Dev.	p10	p50	p90
Experiment A1: 574 participants (81% from the UK, others from the US)					
Age	32.68	10.70	21	30	49
Female	0.41	0.49	0	0	1
Willingness to take risks (1-5)	3.06	0.98	2	3	4
Self-assessed stat. knowledge (1-5)	2.63	1.05	1	3	4
Student	0.19	0.40	0	0	1
Considers regret ex-ante (1-5)	4.04	0.93	3	4	5
Considers regret ex-post (1-5)	3.91	1.02	2	4	5
Time for full experiment (seconds)	434.37	270.63	230	360	690
Experiment A2: 1520 participants (78% from the UK, others from the US)					
Age	36.09	11.86	22	34	53
Female	0.43	0.49	0	0	1
Willingness to take risks (1-5)	3.12	1.06	2	3	4
Self-assessed stat. knowledge (1-5)	2.69	1.08	1	3	4
Student	0.16	0.36	0	0	1
Considers regret ex-ante (1-5)	4.14	0.83	3	4	5
Considers regret ex-post (1-5)	3.89	0.98	2	4	5
Time for full experiment (seconds)	500.14	339.84	234	401	839

Table D.8: Experiment A1 – Choices. In this table, we report average choices in Experiment A1. We report the average fraction of participants selecting each of the assets. See Section B.2 for a more detailed description of the different conditions. t -statistics are based on participant-clustered standard errors whenever both rounds are included. */**/** indicates statistical significance at the 10%/5%/1% level.

Condition	Choice Asset U	Choice Asset F	Difference versus 50%	t -test	N
All					
Sequential Information	27.05%	72.95%	22.95%***	9.81	414
Simultaneous Information Table	49.19%	50.81%	0.81%	0.29	370
Simultaneous Information Verbal	49.73%	50.27%	0.27%	0.09	364
In Levels					
Sequential Information	27.14%	72.86%	22.86%***	7.04	210
Simultaneous Information Table	52.72%	47.28%	-2.72%	-0.68	184
Simultaneous Information Verbal	50.56%	49.44%	-0.56%	-0.13	178
In Differences					
Sequential Information	26.96%	73.04%	23.04%***	6.80	204
Simultaneous Information Table	45.70%	54.30%	4.30%	1.09	186
Simultaneous Information Verbal	48.92%	51.08%	1.08%	0.27	186
Short round (11 outcome pairs)					
Sequential Information	23.19%	76.81%	26.81%***	9.12	207
Simultaneous Information Table	44.32%	55.68%	5.68%	1.55	185
Simultaneous Information Verbal	51.65%	48.35%	-1.65%	-0.44	182
Long round (101 outcome pairs)					
Sequential Information	30.92%	69.08%	19.08%***	5.93	207
Simultaneous Information Table	54.05%	45.95%	-4.05%	-1.10	185
Simultaneous Information Verbal	47.80%	52.20%	2.20%	0.59	182
Round 2 choice, when the round 1 choice was for frequently-outperforming					
Sequential Information	23.18%	76.82%	26.82%***	7.78	151
Simultaneous Information Table	37.50%	62.50%	12.50%**	2.41	88
Simultaneous Information Verbal	40.43%	59.57%	9.57%*	1.88	94
Round 2 choice, when the round 1 choice was for unlikely-outperforming					
Sequential Information	37.50%	62.50%	12.50%*	1.91	56
Simultaneous Information Table	53.61%	46.39%	-3.61%	-0.71	97
Simultaneous Information Verbal	62.50%	37.50%	-12.50%**	-2.41	88

Table D.9: Experiment A1 – Representations. In this table, we report average representations in Experiment A1. Panel A reports the average outcome expected for each of the assets. Panel A reports the average risk score given for each of the assets (from ‘risk-free’ at 1 to ‘very risky’ at 7). See Section B.2 for a more detailed description of the different conditions. *t*-statistics are based on participant-clustered standard errors whenever both rounds are included. */**/** indicates statistical significance at the 10%/5%/1% level.

Condition	Representations Asset U	Representations Asset F	Difference	<i>t</i> -test	<i>N</i>
Panel A: Expected outcome (0-2)					
All					
Sequential Information	0.99	1.14	0.15***	6.04	414
Simultaneous Information Table	1.10	1.05	-0.05	-1.56	370
Simultaneous Information Verbal	0.97	0.95	-0.02	-0.64	364
In Levels					
Sequential Information	0.99	1.15	0.16***	4.87	210
Simultaneous Information Table	1.05	1.02	-0.03	-1.03	184
Simultaneous Information Verbal	0.95	0.95	0.00	0.05	178
In Differences					
Sequential Information	0.99	1.12	0.14***	3.72	204
Simultaneous Information Table	1.14	1.08	-0.06	-1.19	186
Simultaneous Information Verbal	0.98	0.94	-0.05	-0.98	186
Short round (11 outcome pairs)					
Sequential Information	0.94	1.15	0.21***	5.56	207
Simultaneous Information Table	1.10	1.06	-0.04	-1.26	185
Simultaneous Information Verbal	0.96	0.96	0.00	0.05	182
Long round (101 outcome pairs)					
Sequential Information	1.04	1.12	0.09***	3.23	207
Simultaneous Information Table	1.09	1.04	-0.05	-1.20	185
Simultaneous Information Verbal	0.97	0.93	-0.05	-1.05	182
Panel B: Risk assessment (1-7)					
All					
Sequential Information	4.17	3.48	-0.69***	-5.04	414
Simultaneous Information Table	3.81	3.86	0.05	0.48	370
Simultaneous Information Verbal	3.98	4.05	0.07	0.49	364
In Levels					
Sequential Information	4.15	3.36	-0.79***	-4.01	210
Simultaneous Information Table	3.89	3.91	0.02	0.12	184
Simultaneous Information Verbal	3.91	4.06	0.15	0.70	178
In Differences					
Sequential Information	4.20	3.61	-0.59***	-3.09	204
Simultaneous Information Table	3.72	3.81	0.09	0.50	186
Simultaneous Information Verbal	4.04	4.04	0.00	0.00	186
Short round (11 outcome pairs)					
Sequential Information	4.56	3.57	-0.99***	-5.41	207
Simultaneous Information Table	3.94	3.77	-0.17	-1.15	185
Simultaneous Information Verbal	3.92	4.18	0.25	1.27	182
Long round (101 outcome pairs)					
Sequential Information	3.79	3.39	-0.40**	-2.45	207
Simultaneous Information Table	3.68	3.95	0.27*	1.90	185
Simultaneous Information Verbal	4.03	3.93	-0.10	-0.53	182

Table D.10: Experiment A1 – Representations and Choices. In this table, we report average choices in Experiment A1, split based on the representations about the expected outcome and risk of alternatives. We report the average fraction of participants selecting each of the assets. See Section B.2 for a more detailed description of the different conditions. t -statistics are based on participant-clustered standard errors. */**/** indicates statistical significance at the 10%/5%/1% level.

Condition	Choice Asset U	Choice Asset F	Difference versus 50%	t -test	N
Representation in which frequent outperformer has higher expected outcome					
Sequential Information	12.83%	87.17%	37.17%***	17.46	304
Simultaneous Information Table	16.36%	83.64%	33.64%***	9.29	110
Simultaneous Information Verbal	24.31%	75.69%	25.69%***	5.83	144
Representation in which that frequent outperformer has lower expected outcome					
Sequential Information	80.00%	20.00%	-30.00%***	-6.08	70
Simultaneous Information Table	84.76%	15.24%	-34.76%***	-9.69	105
Simultaneous Information Verbal	81.25%	18.75%	-31.25%***	-8.23	112
Representation in which that frequent outperformer has same expected outcome					
Sequential Information	42.50%	57.50%	7.50%	0.51	40
Simultaneous Information Table	48.39%	51.61%	1.61%	0.38	155
Simultaneous Information Verbal	50.93%	49.07%	-0.93%	-0.27	108
Representation in which that frequent outperformer has lower risk					
Sequential Information	13.40%	86.60%	36.60%***	13.28	194
Simultaneous Information Table	26.60%	73.40%	23.40%***	4.99	94
Simultaneous Information Verbal	29.10%	70.90%	20.90%***	4.80	134
Representation in which that frequent outperformer has higher risk					
Sequential Information	46.28%	53.72%	3.72%	0.61	121
Simultaneous Information Table	71.68%	28.32%	-21.68%***	-4.39	113
Simultaneous Information Verbal	69.13%	30.87%	-19.13%***	-4.64	149
Representation in which that frequent outperformer has same risk					
Sequential Information	30.30%	69.70%	19.70%***	4.06	99
Simultaneous Information Table	46.63%	53.37%	3.37%	0.82	163
Simultaneous Information Verbal	48.15%	51.85%	1.85%	0.22	81

Table D.11: Experiment A2 – Choices. In this table, we report average choices in Experiment A2. We report the average fraction of participants selecting each of the assets. See Section B.3 for a more detailed description of the different conditions. t -statistics are based on participant-clustered standard errors whenever both rounds are included.

Condition	Choice Asset U	Choice Asset F	Difference versus 50%	t -test	N	Strength of Preference (1-7)
All						
Sequential Information	24.94%	75.06%	25.06%***	14.98	810	5.42
Simultaneous Information	50.14%	49.86%	-0.14%	-0.08	694	4.44
Without Info Screen						
Sequential Information	23.62%	76.38%	26.38%***	12.05	398	5.58
Simultaneous Information	53.47%	46.53%	-3.47%	-1.27	346	4.54
With Info Screen						
Sequential Information	26.21%	73.79%	23.79%***	9.43	412	5.27
Simultaneous Information	46.84%	53.16%	3.16%	1.18	348	4.34
Short round (11 outcome pairs)						
Sequential Information	22.96%	77.04%	27.04%***	12.92	405	5.29
Simultaneous Information	52.74%	47.26%	-2.74%	-1.02	347	4.39
Long round (101 outcome pairs)						
Sequential Information	26.91%	73.09%	23.09%***	10.46	405	5.56
Simultaneous Information	47.55%	52.45%	2.45%	0.91	347	4.49
Round 2 choice, when the round 1 choice was for frequently-outperforming						
Sequential Information	18.12%	81.88%	31.88%***	14.26	289	5.56
Simultaneous Information	49.71%	50.29%	0.29%	0.08	175	4.54
Round 2 choice, when the round 1 choice was for unlikely-outperforming						
Sequential Information	38.32%	61.68%	11.68%**	2.47	107	5.64
Simultaneous Information	51.74%	48.26%	-1.74%	-0.46	172	4.55

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table D.12: Experiment A2 – Attention to Extremes. In this table, we report regressions of viewing times per outcome pair within the sequential information environment on a dummy for the extreme outcome pair. Viewing time is measured in seconds (Specifications (1) to (3)) or log-seconds (Specifications (4) to (6)). We focus on the short (11-pair) round in Specifications (2) and (5), and on the long (51-pair) round in Specifications (3) and (6). t -statistics are based on participant-clustered standard errors. */**/** indicates statistical significance at the 10%/5%/1% level.

	All	Seconds Short	Long	All	Log(Seconds) Short	Long
$I_{\text{Extreme Pair}}$	1.146***	0.969***	1.325***	0.384***	0.329***	0.438***
	7.53	3.53	8.78	20.17	15.18	14.68
Constant	1.618***	2.967***	1.261***	0.022	0.655***	-0.192***
	16.81	118.99	844.38	1.45	332.99	-650.13
R^2	7.17%	25.41%	5.60%	63.06%	72.13%	60.70%
N	92,078	9,152	82,921	92,078	9,152	82,921
Participant fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Round fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observation number fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Long round fixed effects	Yes	No	No	Yes	No	No
Clustered by participant	Yes	Yes	Yes	Yes	Yes	Yes

Table D.13: Experiment A2 – Choices and Viewing Times. In this table, we test whether longer viewing times for particular outcome pairs are associated with a choice that overweights these outcome pairs in the sequential information environment. We report regressions of a dummy indicating a choice for the frequently-outperforming asset minus 0.5 on a viewing time weighted average outcome difference between the unlikely-outperforming (U) and the frequently-outperforming (F) alternative, based on participant-round specific viewing times per outcome pair. The weighted outcome pair difference is standardized to have a zero mean and standard-deviation of 1 over all participant-rounds. As an illustration, for participants in Specification (1) where all rounds are aggregated, a one standard-deviation increase in the viewing time weighted outperformance of the unlikely-outperforming asset is associated with a 3.9 percentage point lower likelihood to select the frequently-outperforming asset, relative to a baseline likelihood of 75.3% (0.500+0.253). That is, a viewing time weighted performance difference 6.53 (=0.253/0.039) standard-deviation above the average would be required to equalize the likelihood of selecting either asset in the sequential information environment. Notably, the average weighted outcome difference is already positive, because extreme outcome pairs receive overproportional attention on average (see Table D.12), with more than 62% of participants taking more time to view the extreme outcome pair than the average moderate outcome pair. t -statistics are based on participant-clustered standard errors. */**/** indicates statistical significance at the 10%/5%/1% level.

	(1) All	(2) Short	(3) Long
Standardized viewing-time weighted $\overline{X^U} - \overline{X^F}$	-0.039** (-2.47)	-0.013 (-0.63)	-0.061*** (-2.89)
Constant	0.253*** (15.20)	0.271*** (12.93)	0.235*** (10.72)
R ²	0.80%	0.10%	2.04%
N	810	405	405

Table D.14: Experiment A2 – Representations. In this table, we report average representations in Experiment A2. Panel A reports the average outcome expected for each of the assets. Panel B reports the average probability of below/above-median outcomes for each of the alternatives. The t -statistics in Panel B are from a difference-in-difference estimate between the below- and above-median probability assessments. See Section B.3 for a more detailed description of the different conditions. t -statistics are based on participant-clustered standard errors whenever both rounds are included. */**/** indicates statistical significance at the 10%/5%/1% level.

Condition	Representations Asset U	Representations Asset F	Difference (-in-Diff.)	t -test	N
Panel A: Expected outcome (0-2)					
All					
Sequential Information	1.04	1.13	0.08***	5.46	796
Simultaneous Information	1.08	1.05	-0.03	-1.35	740
Without Info Screen					
Sequential Information	1.05	1.14	0.09***	4.31	400
Simultaneous Information	1.08	1.05	-0.03	-0.91	360
With Info Screen					
Sequential Information	1.04	1.11	0.07***	3.38	396
Simultaneous Information	1.08	1.05	-0.03	-1.00	380
Short round (11 outcome pairs)					
Sequential Information	1.03	1.17	0.14***	6.48	398
Simultaneous Information	1.08	1.06	-0.02	-0.78	370
Long round (101 outcome pairs)					
Sequential Information	1.06	1.08	0.02	1.05	398
Simultaneous Information	1.07	1.04	-0.04	-1.19	370
Panel B: Risk assessment (below/above-median probabilities)					
All					
Sequential Information – $P(x < 1)$	44.73%	41.14%	-7.26%***	-6.36	796
Sequential Information – $P(x > 1)$	42.84%	46.50%			
Simultaneous Information – $P(x < 1)$	43.91%	43.94%	0.44%	0.40	740
Simultaneous Information – $P(x > 1)$	40.92%	40.50%			
Without Info Screen					
Sequential Information – $P(x < 1)$	44.07%	40.56%	-6.96%***	-4.57	400
Sequential Information – $P(x > 1)$	43.57%	47.01%			
Simultaneous Information – $P(x < 1)$	44.38%	44.77%	0.66%	0.37	360
Simultaneous Information – $P(x > 1)$	40.13%	39.86%			
With Info Screen					
Sequential Information – $P(x < 1)$	45.39%	41.72%	-7.56%***	-4.28	396
Sequential Information – $P(x > 1)$	42.10%	45.99%			
Simultaneous Information – $P(x < 1)$	43.47%	43.14%	0.24%	0.18	380
Simultaneous Information – $P(x > 1)$	41.67%	41.11%			
Short round (11 outcome pairs)					
Sequential Information – $P(x < 1)$	44.04%	37.93%	-11.69%***	-6.52	398
Sequential Information – $P(x > 1)$	41.50%	47.09%			
Simultaneous Information – $P(x < 1)$	42.37%	42.33%	-0.22%	-0.16	370
Simultaneous Information – $P(x > 1)$	40.33%	40.51%			
Long round (101 outcome pairs)					
Sequential Information – $P(x < 1)$	45.42%	44.34%	-2.83%**	-2.07	398
Sequential Information – $P(x > 1)$	44.18%	45.92%			
Simultaneous Information – $P(x < 1)$	45.46%	45.54%	1.11%	0.70	370
Simultaneous Information – $P(x > 1)$	41.52%	40.49%			

Table D.15: Certainty equivalent minus expected value by condition (all scenes)

Condition	Decoy	Lottery	CE–EV	SE	<i>N</i>
G10	0	0	3.94	0.85	51
G10	1	0	5.11	0.85	45
G10	0	1	3.54	0.85	46
G10	1	1	5.92	0.83	49
G90	0	0	-1.86	0.47	51
G90	1	0	-2.69	0.59	45
G90	0	1	-2.26	0.58	46
G90	1	1	-3.41	0.64	49

Notes: CE–EV is computed from `ce.m.max.prev`, defined as the difference between the estimated certainty equivalent and the expected value of the lottery. *Condition* indexes the probability treatment (G10 = 10% high payoff, G90 = 90% high payoff). *Lottery* indexes the lottery within each probability condition. *Decoy* = 1 indicates the presence of the decoy manipulation; *Decoy* = 0 is the baseline condition.